From Binary to Graded Narrative Identification: Evidence from Tax Policy*

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Abstract

This paper generalizes narrative identification by interpreting documentary classifications as probabilistic information about a latent exogeneity status. Traditional narrative approaches treat selected policy actions as fully exogenous, implicitly imposing a zero-contamination restriction on a chosen subset of observations. In contrast, the framework treats exogeneity as a latent binary attribute and interprets narrative evidence as probabilistic information about whether a given policy action reflects orthogonal variation. Identification exploits variation in narrative support to isolate the effects of policy changes that are most plausibly exogenous, without requiring the construction of observation-level structural shocks. The paper develops a document-grounded implementation that maps policymakers' stated motivations into a continuous index of narrative support for exogeneity while preserving traceability to underlying sources. Empirically, the paper first provides an external validation using a newly assembled dataset of routine U.S. federal tax actions. Narrative-weighted shocks derived from these measures produce macroeconomic responses closely aligned with those obtained from the canonical Romer–Romer series. As an application, the paper constructs a new long-run narrative dataset of French value-added tax policy from 1960 to 2024 and shows that VAT increases with strong narrative support for exogeneity are persistently contractionary, with a short-lived increase in inflation consistent with tax pass-through.

Keywords: Narrative identification; fiscal policy; tax shocks; external instruments; causal inference.
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Introduction

Evaluating the macroeconomic effects of fiscal policy is empirically challenging because policy actions typically respond endogenously to economic conditions. Legislated tax and spending changes reflect contemporaneous shocks, anticipated developments, and evolving policy objectives, confounding causal interpretation. A large literature has proposed identification strategies to address this problem, including structural VARs, external-instrument approaches, and narrative methods that exploit documentary evidence on policymakers' stated motivations. Within this landscape, narrative identification has played a central role in settings where credible instruments are scarce and policy intent must be inferred from historical records. Influential applications span fiscal and monetary policy and include work by Friedman and Schwartz (1963), Mertens and Ravn (2012), Ramey (2011), and Romer and Romer (1989, 2010, 2023). At the same time, narrative approaches remain contested: critics have questioned the credibility of sharp exogeneity classifications, the treatment of mixed motivations, and the sensitivity of results to discretionary inclusion decisions.

Traditional narrative identification maps documentary evidence into a binary inclusion rule. A subset of policy actions is retained when the historical record is judged to support an exogenous interpretation, and retained actions are then treated as fully insulated from endogenous responses. This approach has proved powerful in practice, but it embeds a knife-edge abstraction: once an action is classified as admissible, residual uncertainty about its motivations is ignored. In effect, selected actions are assumed to be free of endogenous contamination, while excluded actions are assumed to contain no usable identifying variation. In practice, however, fiscal measures frequently reflect mixed motivations, and the documentary record often provides only partial or ambiguous support for orthogonality. A binary rule therefore discards information about the intensity and heterogeneity of stated motives and concentrates identification on a small set of extreme cases.

This paper recasts narrative identification through a two latent-type representation that makes explicit the role of classification uncertainty. Rather than asking whether a policy action is exogenous or not, the framework interprets narrative evidence as shifting the probability that a given action belongs to an orthogonal latent class. Intermediate classifications therefore reflect uncertainty about latent exogeneity, not partial exogeneity. Identification proceeds by exploiting systematic variation in narrative support across policy actions. When documentary evidence is informative about how the likelihood of orthogonality varies across actions, the resulting cross-moment variation disciplines the causal interpretation of policy–outcome comovement, without requiring that any observed action be treated as perfectly exogenous. In particular, identification requires that stronger narrative support systematically increases the likelihood that a policy action belongs to the orthogonal latent class, while conditional on latent class membership, narrative support carries no additional information about contemporaneous outcome disturbances.

This perspective shifts the identifying content of narrative analysis away from a zero-contamination requirement for a selected subsample and toward moment restrictions involving imperfect information about latent orthogonality. Causal inference rests on how outcome–policy comovement changes with the strength of narrative support, not on the recovery of observation-level structural shocks. Section 1 formalizes this argument and characterizes the conditions under which variation in narrative support identifies the outcome–policy moment associated with the orthogonal latent class.

To operationalize this perspective, the paper develops a document-grounded implementation that

translates qualitative historical evidence into an empirically usable measure of narrative support for exogeneity. The implementation systematizes the retrieval of relevant policy documents, the organization of stated motivations, and the construction of an index summarizing narrative support. This index is built from *marginal narrative contributions*, which quantify how individual pieces of documentary evidence shift the support for an orthogonal interpretation of a policy action. The procedure preserves traceability to the underlying documentary record and incorporates diagnostics that clarify how the identifying variation used in estimation is supported by the available historical evidence.

The empirical analysis serves two complementary purposes. First, the paper provides an external validation anchored in the canonical narrative tax series of Romer and Romer (2010). Building a new dataset of routine federal tax measures drawn from the annual budget process—covering more than 1,000 policy actions outside the original Romer–Romer sample—the analysis shows that narrative-weighted shocks constructed from these additional measures yield macroeconomic responses that closely align with the benchmark estimates. Second, the paper constructs a new long-run narrative dataset of French value-added tax policy from 1960 to 2024 and estimates the dynamic macroeconomic effects of VAT changes. This application illustrates the value of graded narrative identification in a setting characterized by infrequent but consequential policy interventions and mixed motivations, where binary classification would discard substantial identifying variation.

The paper makes three main contributions. First, it generalizes narrative identification by recasting narrative classifications as probabilistic information about latent exogeneity and by shifting identification toward a moment-based framework that does not require a perfectly exogenous subsample. Second, it develops a disciplined document-based measurement of narrative support that maps policymakers' stated motivations into an empirically usable index while preserving traceability to the underlying sources. Third, it provides external validation and new empirical evidence on the macroeconomic effects of value-added tax changes in France, a central but understudied fiscal instrument.

Literature Review

Identification in Macroeconomics. The paper first situates itself within the literature on policy identification, which includes empirical strategies relying on reduced-form VARs identified through timing, sign, or other model-based restrictions (Blanchard & Perotti, 2002; Sims, 1980; Uhlig, 2005). A distinct line of work instead employs external instruments—variables correlated with the policy shock of interest but orthogonal to other structural innovations—to achieve identification (V. Ramey, 2016; Stock & Watson, 2018). Narrative approaches provide a primary methodology for constructing such instruments, using contemporaneous evidence on policymakers' stated motives. Foundational examples include Friedman and Schwartz (1963) for monetary policy and Hamilton (1985) for oil shocks, with subsequent applications (Ramey, 2011; Ramey & Shapiro, 1998; Ramey & Zubairy, 2018; Romer & Romer, 1989, 2004, 2010) establishing narrative identification as a leading empirical tool. A key limitation of these series is their extreme sparsity: as documented by Barnichon and Mesters (2025), typical narrative shock measures contain roughly 80 percent zeros, which substantially weakens statistical power. Their estimator proposes an econometric remedy for this power–credibility trade-off, but the underlying sparsity reflects, in part, the difficulty of constructing narrative shocks with limited coverage of smaller policy actions.

This paper contributes by reinterpreting narrative classifications as probabilistic information about exogeneity rather than as binary inclusion decisions. Traditional narrative approaches implicitly im-

pose a zero-contamination restriction on a selected subset of policy actions. In contrast, the framework developed here treats exogeneity as a latent property and interprets documentary evidence as graded information about the likelihood that a given action reflects orthogonal policy variation. Identification proceeds through observable cross-moments: variation in narrative support across policy actions isolates the outcome response associated with policy changes that are most plausibly exogenous, without requiring a perfectly exogenous subsample or the recovery of observation-level structural shocks. In this sense, the approach preserves the core logic of narrative identification, using historical evidence to assess orthogonality in expectation, while relaxing the knife-edge assumption that selected actions are free of endogenous contamination.

Text and AI in Economic Analysis. A growing body of work uses text as an empirical input in economics, enabling systematic measurement of beliefs, expectations, and policy intent. Recent surveys formalize core methodological tasks—document similarity, concept detection, relational inference, and text–metadata mapping—and emphasize the importance of transparent links between textual sources and algorithmic outputs (Ash & Hansen, 2023; Gentzkow et al., 2019). The diffusion of large language models (LLMs) has broadened the scope of text-based analysis (Jamilov, 2025; Martell, 2025) but heightened concerns about validation and grounding and its impact on downstream estimated parameters (Battaglia et al., 2024; Carlson & Dell, 2025; Ludwig et al., 2025). This paper contributes to this literature by providing a structured mapping from policy documents to model-generated summaries through semantic retrieval and marginal source contribution, and by adapting a placebo-style validation test from applied econometrics to assess whether model outputs rely on the supplied context.

Recent research also applies text and AI methods to macroeconomic identification. In monetary policy, Aruoba and Drechsel (2025) extract shocks from Federal Reserve documents using natural-language representations, while Hansen and Kazinnik (2023) show that GPT 4 can extract Romer and Romer (2023) monetary policy shocks, reinforcing the feasibility of LLM-based structured classification as used in the final stage of this paper. In fiscal policy, LLMs have been used to detect policy announcements in unstructured text and relate them to high-frequency market reactions (Amodeo & Briganti, 2025; Bandeira et al., 2025). This paper contributes to this strand by developing a framework that, starting from known policy events, extracts and classifies policymakers' stated motives from historical sources, thereby extending narrative identification to large document collections.

Fiscal Policy Transmission and Tax Composition. A final strand of literature concerns the macroeconomic transmission of fiscal policy. Seminal VAR-based analyses (Auerbach & Gorodnichenko, 2012; Blanchard & Perotti, 2002; Mountford & Uhlig, 2009) show that exogenous fiscal expansions raise activity, with multipliers varying across instruments and business-cycle conditions. Narrative studies of government spending shocks (Ramey, 2011; Ramey & Zubairy, 2018) highlight the role of expectations, while analogous work on taxation (Cloyne, 2013; Mertens & Ravn, 2012, 2013; Romer & Romer, 2010) shows that tax multipliers depend on timing, composition, and economic context. Within this broader literature, value-added taxation (VAT) has received far less attention despite its centrality to modern tax systems and its distinctive transmission channel through consumer prices. Existing evidence comes primarily from event and panel studies (Dabla-Norris & Lima, 2023; Riera-Crichton et al., 2016) and micro-level quasi-experiments (Benzarti & Carloni, 2019; Benzarti et al., 2020), which shed light on incidence but provide limited evidence on aggregate dynamics. This paper contributes

by constructing a long-run, single-country narrative series of VAT policy changes for France— the first nation to adopt the modern VAT—allowing for a systematic analysis of its macroeconomic effects.

Latent heterogeneity and imperfect information. The graded narrative framework is conceptually related to econometric approaches that exploit imperfect information about latent states for identification. In the local average treatment effect (LATE) framework, for example, variation in an instrument shifts the probability of belonging to a latent compliance group, allowing identification of a class-specific causal effect without observing latent types directly (Angrist et al., 1996; Imbens & Angrist, 1994). Related work studies settings in which observed indicators provide noisy information about a latent binary condition and derives identification results from restrictions on observable moments (Hu, 2008; Lewbel, 2007; Mahajan, 2006). The present paper adopts this imperfect-information logic in the context of narrative identification. It does not seek to recover latent classifications or impose a specific functional form on the relationship between narrative support and outcome–policy comovement. Instead, it exploits systematic variation in documentary support to learn about the structural effect associated with policy changes that are most plausibly orthogonal.

The remainder of the paper is structured as follows. Section 1 develops the conceptual framework for narrative identification and formalizes the graded, moment-based interpretation of policy shocks. Section 2 describes the document-grounded implementation that operationalizes this framework by translating narrative evidence into an empirically usable object. Section 3 introduces the marginal narrative contribution procedure and associated placebo-based diagnostics used to assess textual grounding.

The empirical analysis is organized by country. Section 4 presents the French application, including the construction of the VAT dataset and the underlying textual corpus, and estimates the macroeconomic effects of narrative-weighted VAT shocks. Section 5 turns to the United States, documents the construction of the federal tax dataset and associated policy documents, and provides an external validation by replicating and extending Romer and Romer (2010). The final section concludes.

1 Mathematical Model for Narrative Identification

The objective of this paper is to estimate the causal effects of legislated tax changes on macroeconomic outcomes using narrative evidence from policy documents. Let y_t denote an aggregate outcome of interest. In each period t a set of legislated policy actions indexed by j is enacted, with action-level changes $\Delta\tau_{j,t}$ and aggregate change

$$\Delta\tau_t \;=\; \sum_j \Delta\tau_{j,t}.$$

For each horizon $h \geq 0$, consider the structural relation

$$y_{t+h} = \beta_h \Delta\tau_t + \varepsilon_{t+h}, \tag{1}$$

where β_h denotes the dynamic causal effect of an orthogonal tax innovation. Identification of β_h requires policy variation orthogonal to ε_{t+h}, which typically fails because tax policy responds endogenously to contemporaneous or anticipated economic conditions.

Narrative identification proceeds at the level of action (j,t): for each policy action, documentary sources provide evidence about the motivations behind it, and that evidence varies in how strongly it supports an orthogonal interpretation. This section formalizes how graded narrative support can be used for identification and how it maps into estimands in the local-projection implementation used later in the paper. The argument proceeds in three steps. Section 1.1 shows how the traditional binary approach can be read as a limiting case of the graded framework, making explicit the restrictions it silently imposes. Section 1.2 introduces a two-types representation that provides a structural interpretation of $p_{j,t}$ and motivates the affine structure underlying the two-types-LP estimator. Section 1.3 shows that the core identification result rests on a moment restriction weaker than the two-types model, a property that motivates the menu of estimators in Section 1.4. Formal derivations are collected in Appendix A.

1.1 Traditional Narrative Identification

Traditional narrative identification maps documentary evidence into a binary inclusion decision. From the population of legislated tax changes, a subset of policy actions is selected and treated as providing identifying variation, while remaining actions are discarded. Represent this decision by an indicator $\mathbb{1}\{\Delta\tau_{j,t} \text{ is classified as exogenous}\}$, equal to one when the action is retained and zero otherwise.

Although traditional studies do not report an explicit probability, their decision rule can be represented as thresholding an implicit document-based assessment $p_{j,t} \in [0,1]$:

$$\mathbb{1}\{\Delta\tau_{j,t} \text{ is classified as exogenous}\} \;=\; \mathbb{1}\{p_{j,t} \geq \bar{p}\},$$

for some evidentiary cutoff $\bar{p} \in (0,1)$. Once the cutoff is applied, residual classification uncertainty is ignored: included actions are treated as providing orthogonal policy variation. This approach embeds two restrictions. First, the binary inclusion rule discards all identifying information from actions with intermediate narrative support. Second, the zero-contamination restriction treats retained actions as perfectly exogenous, ignoring residual uncertainty about their motivations. This paper relaxes both restrictions simultaneously.

1.2 A two-types Representation of Fiscal Policy Actions

To give precise content to what graded narrative support identifies, this subsection introduces a two-types two-types representation of fiscal policy actions. The representation serves two purposes: it provides a structural interpretation of the narrative support index $p_{j,t}$ as the probability that an action belongs to an orthogonal latent class, and it motivates the affine structure of $\mu_h(p)$ that underlies the two-types-LP estimator.

A two-types two-types model. For a given policy action (j,t), suppose the observed legislated change can be written as

$$\Delta\tau_{j,t} \;=\; u_{j,t}\, a_{j,t} + (1 - u_{j,t})\, b_{j,t}, \qquad u_{j,t} \in \{0,1\}. \tag{2}$$

When $u_{j,t} = 1$, the action loads on an *orthogonal* source $a_{j,t}$; when $u_{j,t} = 0$, it loads on a *contaminated* source $b_{j,t}$, which may be correlated with macroeconomic disturbances. The orthogonal component sat-

isfies $\mathbb{E}(a_{j,t}\,\varepsilon_{t+h}) = 0$. This structure captures the idea that fiscal actions reflect heterogeneous sources of variation, some driven by long-run structural or ideological considerations plausibly orthogonal to short-run conditions, others driven by cyclical or stabilization motives that are not. Mixed-motive actions, where both rationales coexist, are accommodated by interpreting $u_{j,t}$ as selecting the dominant motive.

Narrative evidence as graded support. Under the two-types model, the narrative support index $p_{j,t} \in [0,1]$ is interpreted as

$$p_{j,t} \;=\; \Pr(u_{j,t} = 1 \mid \mathcal{D}_{j,t}),$$

the posterior probability of belonging to the orthogonal class given the documentary record $\mathcal{D}_{j,t}$. Intermediate values of $p_{j,t}$ therefore reflect classification uncertainty induced by an incomplete or ambiguous historical record, not partial exogeneity. Traditional binary identification is the special case in which documentary evidence is treated as observed with certainty: retained actions are coded as $p_{j,t} = 1$ and discarded actions as $p_{j,t} = 0$.

Implication for the support-indexed cross-moment. If $p_{j,t}$ is a sufficient statistic for $u_{j,t}$ conditional on outcomes, iterated expectations over (2) yield

$$\mu_h(p) \;=\; p\,\mathbb{E}(y_{t+h}\,a_{j,t}) + (1-p)\,\mathbb{E}(y_{t+h}\,b_{j,t}), \tag{3}$$

so $\mu_h(p)$ is affine in p. This affine structure is the model-based justification for the two-types-LP estimator. As established in the next subsection, identification of β_h does not require (3) to hold exactly: the boundary orthogonality condition restricts only the limit of $\mu_h(p)$ as $p \to 1$, not its interior shape.

1.3 Moment-Based Identification and a Boundary Estimand

This subsection connects graded narrative support to an estimand that can be recovered from observable moments without requiring the full two-types structure of Section 1.2.

Support-indexed moments and the boundary object. Define the support-indexed cross-moment and second moment

$$\mu_h(p) \;\equiv\; \mathbb{E}(y_{t+h}\,\Delta\tau_{j,t} \mid p_{j,t} = p)\,, \qquad v(p) \;\equiv\; \mathbb{E}\big(\Delta\tau_{j,t}^2 \mid p_{j,t} = p\big)\,. \tag{4}$$

When $v(p) > 0$, define the support-indexed response $r_h(p) \equiv \mu_h(p)/v(p)$. Using (1),

$$\mu_h(p) \;=\; \beta_h\,v(p) + \mathbb{E}(\varepsilon_{t+h}\,\Delta\tau_{j,t} \mid p_{j,t} = p)\,,$$

so $r_h(p)$ equals β_h up to the support-indexed endogeneity term.

Assumption (Boundary orthogonality).

$$\lim_{p\to 1}\; \mathbb{E}(\varepsilon_{t+h}\,\Delta\tau_{j,t} \mid p_{j,t} = p) = 0. \tag{5}$$

As narrative support approaches its upper limit, the conditional covariance between policy changes and macroeconomic disturbances converges to zero. Under continuity of the relevant conditional moments and sufficient support of $p_{j,t}$ near 1, the causal object of interest is the boundary response $r_h(1) \equiv \lim_{p\to 1} r_h(p)$, which coincides with β_h. Appendix A.1 provides the formal argument.

Remark 1.1 (Relationship to traditional narrative identification). Assumption (5) is a boundary weakening of the zero-contamination restriction implicit in traditional narrative designs, which assert $\mathbb{E}(\varepsilon_{t+h}\,\Delta\tau_{j,t} \mid p_{j,t} \geq \bar{p}) = 0$ globally for all retained actions. The present assumption requires this orthogonality to hold only in the limit as $p_{j,t} \to 1$, permitting nonzero endogeneity for actions with intermediate support. Traditional binary identification is recovered as the special case in which the boundary condition collapses to a global restriction.

Remark 1.2 (Generality relative to the two-types model). Assumption (5) is weaker than the two-types representation of Section 1.2. Under the two-types model, boundary orthogonality follows from $\mathbb{E}(a_{j,t}\,\varepsilon_{t+h}) = 0$: as $p_{j,t} \to 1$, the action is almost surely drawn from the orthogonal class. The converse does not hold: boundary orthogonality places no restriction on the shape of $\mu_h(p)$ in the interior of $[0, 1)$ and does not require the affine structure (3). Identification of β_h is therefore robust to misspecification of the two-types model, while the efficiency of estimation depends on how well the chosen estimator captures the shape of $r_h(p)$.

1.4 Econometric Implementation

This subsection translates the identification result into a set of operational estimators, each making a different assumption about the shape of $r_h(p)$ and thereby occupying a different position on the spectrum between robustness and efficiency.

From action-level support to time-t policy measures. Let $g : [0, 1] \to \mathbb{R}$ be a weighting rule. Given action-level narrative support $p_{j,t}$, define the time-t policy measure

$$s_t(g) \;\equiv\; \sum_j g(p_{j,t})\,\Delta\tau_{j,t}. \tag{6}$$

A central special case is the hard-threshold rule $g(p) = \mathbb{1}\{p \geq \bar{p}\}$, which reproduces traditional narrative identification. Smooth weights such as $g(p) = p$ or $g(p) = p^\kappa$ extend this by placing increasing emphasis on high-support actions.

Three estimators as competing strategies. The choice of weighting rule g is not merely a technical convenience — it reflects a substantive choice about how to approach the boundary estimand $r_h(1)$ given uncertainty about the shape of $r_h(p)$ in the interior. Three natural strategies emerge, each occupying a different position along the spectrum between robustness and efficiency. The formal decomposition underlying this tradeoff is in Appendix A.3.

Hard thresholding, $g(p) = \mathbb{1}\{p \geq \bar{p}\}$, concentrates all identifying weight on actions with narrative support above the cutoff $\bar{p}$. As $\bar{p} \to 1$, the estimand approaches $r_h(1)$ under boundary orthogonality alone, with no assumption on the interior shape of $r_h(p)$. This makes hard thresholding the most robust strategy, but it discards identifying variation from intermediate-support actions and becomes imprecise as the effective sample shrinks.

Soft weighting, $g(p) = p$ or $g(p) = p^\kappa$, retains the full sample while downweighting low-support actions. This typically improves precision relative to hard thresholding, but the resulting coefficient identifies a support-weighted average of $r_h(p)$ over the interior of the distribution rather than the boundary object. The proximity of this average to $r_h(1)$ depends on the shape of $r_h(p)$, which is unknown without additional assumptions.

two-types-LP, described below, parameterizes $r_h(p)$ as affine in p and evaluates the fitted function at $p = 1$. This strategy is motivated directly by the two-types model of Section 1.2: if the affine structure (3) holds, the two-types-LP delivers a consistent and more efficient estimate of $r_h(1)$ than hard thresholding by using information from the entire distribution of $p_{j,t}$. If the affine structure is misspecified, the two-types-LP boundary estimate inherits that misspecification, while hard thresholding remains consistent under boundary orthogonality alone. The choice among these three strategies therefore reflects the researcher's confidence in the two-types model: the two-types-LP is preferred when the model is trusted, hard thresholding when robustness to model misspecification is the priority, and the nesting diagnostic described below provides empirical guidance on which regime applies in a given application.

Weighted local projections (reweighting estimands). A natural baseline regresses y_{t+h} on the proxy $s_t(g)$:

$$y_{t+h} \;=\; \alpha_h + \beta_h(g)\, s_t(g) + \Gamma_h' X_t + u_{t+h}, \tag{7}$$

where X_t collects standard controls. Under standard regularity conditions, $\beta_h(g)$ identifies a reweighted average of $r_h(p)$ over the empirical distribution of $p_{j,t}$. A closed-form expression for the proxy-LP estimand is in Appendix A.2.

Two-types local projections (boundary extrapolation). To target $r_h(1)$ directly, the two-types model motivates the affine approximation $r_h(p) \approx \theta_h + \lambda_h p$, which corresponds to the *two-types* local projection

$$y_{t+h} \;=\; \alpha_h + \theta_h\, \Delta\tau_t + \lambda_h\, s_t(p) + \Gamma_h' X_t + u_{t+h}, \tag{8}$$

where $s_t(p) = \sum_j p_{j,t} \Delta\tau_{j,t}$. The implied boundary estimate is

$$\widehat{r}_h(1) \;\equiv\; \widehat{\theta}_h + \widehat{\lambda}_h. \tag{9}$$

A nesting diagnostic. The proxy LP using $s_t(p)$ is nested in (8) as the restriction $\theta_h = 0$. Testing $\theta_h = 0$ provides a diagnostic of whether compressing graded information into the single proxy is adequate at horizon h. Comparing two-types-LP boundary estimates with hard-threshold estimates at high $\bar{p}$ provides an additional robustness check on the affine extrapolation: agreement between the two approaches indicates that results are not driven by the parameterization of $r_h(p)$ in the interior.

Assumptions roadmap. The econometric implementation separates three layers of assumptions: (i) the core identifying restriction is boundary orthogonality (5); (ii) learning about $r_h(1)$ additionally requires adequate support of $p_{j,t}$ near 1; and (iii) the two-types local projection (8) adds an approximation assumption about the shape of $r_h(p)$, affine under the two-types model. The empirical section reports proxy LP estimates under alternative weighting rules, two-types-LP boundary estimates with

the nesting diagnostic, and hard-threshold estimates as a robustness check, providing transparent evidence on how results depend on both the use of graded narrative support and the choice of estimation strategy.

2 A Document-Grounded Implementation of Graded Narrative Identification

Section 1 establishes that identification exploits variation in the narrative support index $p_{j,t}$, interpreted as probabilistic information about latent exogeneity. This section describes how contemporaneous policy documents are mapped into observable narrative classifications that underpin the construction of that index. For expositional convenience, I refer to this document-grounded implementation as SANI. The emphasis, however, is on the economic mapping from textual evidence to narrative classifications rather than on the procedure itself.

2.1 From policy documents to narrative classifications

Let j index individual policy events within period t. Each event (j, t) is associated with (i) a short policy description $q_{j,t}$ summarizing the measure as recorded in fiscal documents, and (ii) a contemporaneous document collection $\mathcal{D}_t$ containing texts relevant for identifying the motivations behind policy changes—for example, budget reports or legislative debates.[1] The collection is partitioned into N text segments $d_{t,k}$ of length L,

$$\mathcal{D}_t = \{d_{t,1}, \ldots, d_{t,N}\}.$$

For brevity, the time index is omitted in what follows. A maintained assumption is that the contemporaneous corpus $\mathcal{D}_t$ contains the relevant statements of motivation for each policy event.

To fix ideas, consider a representative case: the major reform of the standard value-added tax (VAT) rate proposed in the 1977 French budget. The policy description q_j recorded in the fiscal documents reads: *"reduction of the standard rate and unification of the standard rate and the intermediate rate."* The measure's reported fiscal impact is 8.5 billion francs (0.5% of GDP). This example will be used throughout to illustrate how the formal operators map raw documents into narrative classifications.

Selection of relevant documentary evidence. The operator $\mathcal{R}_{K,L}$ selects, from the contemporaneous corpus $\mathcal{D}$, the text segments most closely related to the policy description q_j. An embedding model maps q_j and each segment d_k into a shared vector space, and relevance is measured by the distance

$$s_{j,k} = \mathrm{dist}(q_j, d_k), \qquad d_k \in \mathcal{D}.$$

The embedding model is a pretrained transformer-based neural network that maps text into fixed-dimensional vector representations. The operator returns the K closest segments:

$$\mathcal{R}_{K,L}(q_j; \mathcal{D}) = \underset{d_k \in \mathcal{D}}{\overset{K}{\arg\min}}\ s_{j,k} = \{d^{(1)}, \ldots, d^{(K)}\}, \tag{10}$$

[1] In many applications, documents cover multiple measures within a period, making it natural to define a common corpus $\mathcal{D}_t$. When measure-specific documents are available, the framework can instead define a separate corpus $\mathcal{D}_{j,t}$ for each event.

where $d^{(k)}$ denotes the k-th smallest-distance segment.[2] The choice of segment length L and the number of retrieved segments K governs the scope of the selected evidence; Section 4.2.2 documents the calibration used in the application.

In the 1977 VAT example, the corpus $\mathcal{D}_t$ consists of the full Senate report on the budget, partitioned into approximately 800 segments of $L = 2{,}000$ characters. Applying $\mathcal{R}_{K,L}$ with $K = 10$ yields a set of passages whose first three correspond directly to the article of law implementing the VAT reform and together account for more than 80% of its text; the remaining segments relate to other fiscal measures. The first five retrieved segments are reported in Appendix B.3. This step isolates the contemporaneous statements most likely to contain the government's stated motivations.

Structured summarisation. The second stage condenses the retrieved set $\hat{R}_j$ into a structured narrative $\hat{S}_j$ that records the policy change and its stated motivations:

$$\mathcal{S}(\hat{R}_j) = \hat{S}_j.$$

The objective is compression rather than reinterpretation: the operator extracts and organizes motivations expressed in the retrieved passages while remaining faithful to their content. In practice, $\mathcal{S}$ is implemented using a pretrained large-scale transformer language model trained on general text corpora.

Applied to the ten retrieved passages for the 1977 VAT reform, the summarisation operator produces the following structured narrative $\hat{S}_j$ (verbatim model output in italics):

- ***Mitigate the rise in prices after the end of the price freeze****: The measure aims to limit the inflationary rebound likely to occur at the end of the price-freeze period (December 31, 1976) by reducing the tax burden on consumption.*
- ***Complete a structural reform****: This change represents the final step in a reform undertaken since 1969, aimed at merging the standard VAT rate with the intermediate rate and reducing the number of VAT rates.*
- ***Fight inflation****: The reduction of the rate is part of the broader government programme to combat inflation by easing indirect taxation.*

These motivations capture the full substantive content of the official justification. Two relate to short-run stabilisation concerns, while the second reflects a longer-run structural objective.

Classification. The final stage maps the structured narrative $\hat{S}_j$ into a set of application-specific categories using category-specific prompts. Formally,

$$\mathcal{C}(\hat{S}_j) = \hat{c}_j \in \mathcal{G},$$

where $\mathcal{G}$ denotes the set of admissible categories—for example, $\mathcal{G} = \{\text{exogenous}, \text{endogenous}\}$ when studying tax policy. Each category is queried independently, yielding a binary classification together with a concise rationale explicitly tied to the structured narrative $\hat{S}_j$. This modular design isolates

[2] The threshold formulation $s_j^{(k)} \leq \max_{d \in \hat{R}_j} s_{j,d}$ is equivalent.

classification errors within a single category and allows the set $\mathcal{G}$ to be refined or extended as needed. When mutually exclusive classifications are required, explicit resolution rules can be imposed.

In the 1977 VAT example, the first and third motivations are classified as *Demand Management*, reflecting their explicit inflation-related objectives, while the second is classified as *Long-Term Structural Reform*. A single policy measure thus combines short-run macroeconomic aims with longer-run structural goals.

This feature is central to the graded interpretation developed in Section 1. Traditional narrative approaches typically resolve such cases by selecting a dominant motivation (Romer & Romer, 2010) or presenting alternative classifications as robustness checks (Cloyne, 2013). Here, all stated motives are recorded and classified. Narrative evidence may therefore support both orthogonal and non-orthogonal rationales within a single policy action, generating intermediate degrees of narrative support rather than a binary inclusion decision. This structure allows downstream analyses to incorporate mixed motivations directly, for instance through exogeneity shares.

Reproducibility. All prompting and decoding parameters are collected in a fixed vector θ, with stochastic elements of text generation disabled (temperature set to zero; see Appendix B.4). Under these settings, the retrieval, summarisation, and classification operators are deterministic, i.e., $\mathcal{S}(\cdot;\theta) = \mathcal{S}(\cdot)$. Open-source models ensure perfect reproducibility; proprietary models may be subject to silent updates, and researchers should therefore log metadata such as model version and timestamp.

End-to-end mapping. Together, these steps define the mapping

$$(q_{j,t}, \mathcal{D}_t) \xrightarrow{\mathcal{R}_{K,L}} \hat{R}_{j,t} \xrightarrow{\mathcal{S}} \hat{S}_{j,t} \xrightarrow{\mathcal{C}} \hat{c}_{j,t},$$

which transforms each policy event into a structured narrative and its associated classifications. These classifications form the empirical basis for constructing the narrative support index $q_{j,t}$ used in the graded identification framework. A schematic illustration of this mapping is provided in Appendix Figure 10.

2.2 From Category Assignments to Narrative Support Probabilities

The classification stage maps each structured motivation M_ℓ into category-specific outputs for categories $c \in \mathcal{G}$. These outputs are informative, but they should not be interpreted directly as probabilities over the true narrative classes. Instead, I treat them as noisy measurements of an underlying latent category and use a multinomial measurement model, estimated on a calibration sample with labeled categories, to recover class probabilities. These class probabilities are then aggregated into the probability that a motivation is exogenous.

Category-specific signals as measurements. Fix one implementation of the retrieval, summarisation, and classification procedure. For each motivation M_ℓ and category $c \in \mathcal{G}$, the classification operator produces a signal $s_{\ell,c}$ measuring the degree of documentary support for assigning M_ℓ to category c. One possible implementation is

$$s_{\ell,c} = \log \Pr_{\text{LM}}(\text{Yes} \mid X_c \cup M_\ell) - \log \Pr_{\text{LM}}(\text{Yes} \mid X_c), \tag{11}$$

where X_c denotes the category-specific prompt and $X_c \cup M_\ell$ augments that prompt with motivation M_ℓ. The term $s_{\ell,c}$ therefore captures the incremental evidence, relative to the prompt-specific baseline, that the motivation supports category c.

Collect these measurements in the signal vector

$$Z_\ell = (s_{\ell,c})_{c \in \mathcal{G}}.$$

Because the elements of Z_ℓ are generated by separate prompts and need not be comparable on a common probability scale, they are treated as noisy measurements of the underlying category rather than as class probabilities themselves.

A multinomial measurement model. Let $C_\ell \in \mathcal{G}$ denote the latent narrative category of motivation M_ℓ. On a calibration sample for which C_ℓ is observed from external human coding, I estimate a multinomial logit model linking the true category to the signal vector Z_ℓ:

$$\Pr(C_\ell = c \mid Z_\ell) = \frac{\exp(\alpha_c + Z_\ell' \beta_c)}{\sum_{c' \in \mathcal{G}} \exp(\alpha_{c'} + Z_\ell' \beta_{c'})}, \qquad c \in \mathcal{G}. \tag{12}$$

Here, α_c is a category-specific intercept and β_c captures how the full vector of measured signals predicts assignment to category c. This specification allows the calibration data to correct for systematic distortions in the raw model outputs, including differences in scale, prompt-specific bias, and cross-category confusion.

Estimated probabilities from equation (12) are denoted

$$\hat{P}_{\ell,c} = \hat{\Pr}(C_\ell = c \mid Z_\ell).$$

These are interpreted as calibrated probabilities that motivation M_ℓ belongs to category c, conditional on the measured signal vector produced by the language-model procedure.

From class probabilities to exogeneity probabilities. Let $\mathcal{G}_{\text{exo}} \subset \mathcal{G}$ denote the subset of categories classified as exogenous in the sense of Section 1, and let $\mathcal{G}_{\text{endo}} = \mathcal{G} \setminus \mathcal{G}_{\text{exo}}$ denote the complementary endogenous set. The motivation-level probability of exogeneity is then defined as

$$p_\ell = \Pr(Y_\ell = 1 \mid Z_\ell) = \sum_{c \in \mathcal{G}_{\text{exo}}} \Pr(C_\ell = c \mid Z_\ell), \tag{13}$$

with empirical counterpart

$$\hat{p}_\ell = \sum_{c \in \mathcal{G}_{\text{exo}}} \hat{P}_{\ell,c}, \tag{14}$$

where $Y_\ell = 1$ indicates that motivation M_ℓ is exogenous. Thus, the exogeneity probability is obtained by summing calibrated class probabilities over all categories belonging to the exogenous set.

This formulation preserves the graded nature of the narrative evidence. A motivation may load on several categories simultaneously through the measurement vector Z_ℓ, while the multinomial model converts these noisy signals into a coherent probability distribution over the true taxonomy.

Measure-level aggregation. A policy measure may contain several stated motivations. Let $\mathcal{M}_{j,t}$ denote the set of motivations extracted for measure (j,t). The narrative support probability entering the identification framework is defined as the average motivation-level exogeneity probability:

$$p_{j,t} = \frac{1}{|\mathcal{M}_{j,t}|} \sum_{\ell \in \mathcal{M}_{j,t}} \hat{p}_\ell. \tag{15}$$

Under equal weighting, each motivation contributes equally to the exogeneity content of the policy action. The resulting object $p_{j,t} \in [0,1]$ summarizes the extent to which the contemporaneous documentary record supports an exogenous interpretation of measure (j,t).

Interpretation. This construction treats category-specific language-model outputs as measurements rather than direct observations of latent narrative classes. The multinomial logit in equation (12) calibrates these measured signals to hand-coded categories, while equations (13)–(15) map the calibrated class probabilities into the scalar probability object required by the graded identification framework. Accordingly, $p_{j,t}$ should be interpreted as a calibrated, within-procedure estimate of narrative exogeneity rather than as an error-free quantity.

VAT illustrative example. In the 1977 VAT reform example, the extracted motivations generate a vector of category-specific signals Z_ℓ across the narrative taxonomy. Feeding these signals into the estimated multinomial measurement model yields calibrated probabilities $\hat{P}_{\ell,c}$ for each category. Summing over the categories classified as exogenous produces the motivation-level exogeneity probabilities $\hat{p}_\ell$, and averaging these across motivations gives the policy-level narrative support probability $p_{j,t}$.

Table 1: Estimated marginal narrative contribution across Cloyne (2013) classification categories

	Category	**M1**	**M2**	**M3**
Endo	Demand Management (DM)	0.55	0.31	0.38
	Spending-Driven (SD)	0.00	0.00	0.28
	Supply Stimulus (SS)	0.00	0.10	0.00
	Deficit Reduction (DR)	0.03	0.00	0.21
Exo	Deficit Consolidation (DC)	0.17	0.23	0.00
	External (EXT)	0.34	0.21	0.22
	Ideological (IL)	0.35	0.39	0.14
	Long-Run (LR)	0.21	0.67	0.30

Note: Entries report the *calibrated* marginal narrative contribution of each motivation to the corresponding classification category. Larger values indicate stronger documentary support for assigning the measure to that category. For each motivation, the shaded cell marks the category receiving the highest contribution, indicating the most strongly supported interpretation. "Endo" denotes endogenous subcategories and "Exo" denotes exogenous subcategories as defined in Cloyne (2013). Detailed category definitions appear in Table 1.

3 Measurement and Grounding Diagnostics

Section 1 defines identification in terms of variation in the narrative support index $p_{j,t}$. Section 2.2 describes how this index is constructed from category-specific signals extracted from contemporaneous policy documents and calibrated into exogeneity probabilities. Because identification relies on

these quantities, it is essential to assess both their stability across admissible implementations of the procedure and their grounding in the underlying documentary evidence.

Two concerns arise naturally. First, alternative implementations of the retrieval, summarisation, and classification procedure may generate different measurements of narrative support, even when applied to the same policy event and the same document collection. Second, model outputs may reflect generic linguistic regularities or pretraining priors rather than the policy-specific textual evidence supplied to the model. If either concern is severe, the resulting narrative support index may fail to capture the identifying variation of interest.

This section develops diagnostics for both sources of uncertainty. It first studies measurement dispersion across implementations of the SANI procedure and shows how alternative implementation-specific estimates can be interpreted as noisy measurements of a common latent narrative object. It then develops grounding diagnostics that quantify the dependence of model outputs on the retrieved documentary record.

The key grounding object is the *marginal narrative contribution*: the change in the model's output induced by withholding a specific unit of textual evidence while holding all other inputs fixed. Whereas Section 2.2 establishes the baseline mapping from extracted motivations to narrative support probabilities, the present section examines whether that mapping is stable across implementations and whether generated motivations are anchored in specific retrieved passages.

The grounding analysis is related to a growing literature studying the sensitivity of large language model outputs to input perturbations (e.g., Cohen-Wang et al. (2024), Feng et al. (2018), and Prabhakaran et al. (2019)). Unlike token- or sentence-level attribution methods, the present framework operates at the level of economically meaningful evidence units—retrieved documentary passages—and is tailored to assessing both measurement stability and documentary grounding in narrative identification.

3.1 Measurement Dispersion Across Implementations

The probability construction in Section 2.2 is defined for a single implementation of the retrieval, summarisation, and classification procedure. In practice, however, multiple admissible implementations may be used. These implementations may differ because they rely on different base language models, different chunking rules, different retrieval depths, different summarisation settings, or different category prompts. Such variation can affect not only category assignments, but also the number, ordering, and wording of extracted motivations. For this reason, the stable object of interest is not the motivation-level classification itself, but the policy-event-level narrative support for exogeneity.

A latent policy-level narrative object. Let $p^{\dagger}_{j,t} \in [0,1]$ denote the latent narrative support for exogeneity associated with policy event (j,t). This object is interpreted as the probability, conditional on the full contemporaneous documentary record, that the policy change reflects exogenous variation in the sense of Section 1. It is not directly observed.

Each implementation $r \in \mathcal{R}$ applies the document-based procedure to the same policy event (j,t) and produces a policy-level estimate

$$\hat{p}^{(r)}_{j,t} \in [0,1].$$

This estimate may be constructed through different intermediate summaries, different extracted mo-

tivations, or different category assignments. I therefore treat the implementation-specific quantities $\{\hat{p}_{j,t}^{(r)}\}_{r\in\mathcal{R}}$ as alternative measurements of the same latent policy-level object $p_{j,t}^{\dagger}$.

A measurement-error representation. Formally, suppose that each implementation satisfies

$$\hat{p}_{j,t}^{(r)} = p_{j,t}^{\dagger} + \varepsilon_{j,t}^{(r)}, \qquad r \in \mathcal{R}, \tag{16}$$

where $\varepsilon_{j,t}^{(r)}$ captures implementation-specific measurement error arising from model choice, retrieval design, summarisation differences, prompt wording, and related specification choices.

This representation makes clear that dispersion across implementations does not correspond to different economic objects. Rather, each implementation provides an imperfect measurement of a common latent quantity: the narrative support for exogeneity of policy event (j, t).

Aggregation across implementations. Under the maintained assumption that implementation-specific errors are mean-zero conditional on the latent signal,

$$\mathbb{E}\Big[\varepsilon_{j,t}^{(r)} \mid p_{j,t}^{\dagger}\Big] = 0, \tag{17}$$

a natural estimator of $p_{j,t}^{\dagger}$ is the average across implementations:

$$\bar{p}_{j,t} = \frac{1}{|\mathcal{R}|} \sum_{r\in\mathcal{R}} \hat{p}_{j,t}^{(r)}. \tag{18}$$

This aggregation reduces idiosyncratic implementation noise and isolates the component of the narrative signal that is common across admissible specifications.

More generally, one could consider a weighted average,

$$\bar{p}_{j,t} = \sum_{r\in\mathcal{R}} \omega_r \hat{p}_{j,t}^{(r)}, \qquad \sum_{r\in\mathcal{R}} \omega_r = 1, \tag{19}$$

where ω_r reflects implementation-specific precision. In the baseline specification, I use equal weights in order to avoid imposing additional structure on the measurement system.

Interpretation and use in the empirical analysis. Equation (16) defines a classical multiple-measurement setting in which several implementation-specific proxies are observed for the same latent policy-level object $p_{j,t}^{\dagger}$. Because intermediate objects, such as extracted motivations, are themselves implementation-dependent, no single specification provides a privileged mapping from documents to narrative support. The cross-implementation average $\bar{p}_{j,t}$ therefore isolates the component of the signal that is common across admissible implementations while reducing idiosyncratic measurement noise.

In the empirical analysis, I use $\bar{p}_{j,t}$ as the measure of narrative support entering the identification framework. Accordingly, the results should be interpreted as exploiting variation in a noisy proxy for $p_{j,t}^{\dagger}$. Under the maintained assumption that implementation-specific errors are mean-zero and not perfectly correlated across specifications, averaging reduces variance without introducing systematic bias in the identifying variation.

3.2 Documentary Grounding

The previous subsection examines dispersion across implementations and interprets it as measurement error in a common latent policy-level object. I now turn to a distinct concern: whether the measured narrative support is grounded in the contemporaneous documentary evidence.

Even if different implementations produce similar values of $p_{j,t}$, this does not guarantee that these values are driven by the policy-specific documents supplied to the model. Model outputs may instead reflect generic linguistic regularities or pretraining priors. To assess grounding, I study the sensitivity of the model's output to systematic perturbations of the retrieved evidence.

The approach builds on a broader literature on perturbation-based analysis of model behavior (Cohen-Wang et al., 2024; Feng et al., 2018; Prabhakaran et al., 2019). Unlike token- or sentence-level attribution methods, the present framework operates at the level of economically meaningful evidence units—retrieved documentary passages—and is tailored to the narrative identification setting.

3.2.1 Marginal Contributions

For a given policy measure j, the retrieval operator produces a set of passages

$$\hat{R}_j = \{d_1, \ldots, d_K\},$$

which jointly determine the structured narrative $\hat{S}_j$ and the resulting narrative support measure. Grounding requires assessing whether these outputs depend substantively on specific elements of $\hat{R}_j$.

To do so, I consider counterfactual evaluations in which subsets of retrieved passages are withheld while holding all other inputs fixed. If a given passage contains information that is relevant for the policy's stated motivations, removing it should affect the model's output. Conversely, passages that are unrelated to the policy should have little effect.

Inclusion design. Introduce an inclusion vector

$$v \in \{0, 1\}^K,$$

where $v_\ell = 1$ indicates inclusion of passage d_ℓ and $v_\ell = 0$ indicates that it is withheld. Let

$$g(v)$$

denote the log-odds of the model output of interest under inclusion pattern v, holding the prompting and decoding parameters fixed.

The marginal contribution of passage d_ℓ at context $v_{-\ell}$ is defined as

$$\Delta_\ell^g(v_{-\ell}) = g(v_\ell = 1, v_{-\ell}) - g(v_\ell = 0, v_{-\ell}),$$

which measures the change in output induced by including passage d_ℓ.

Linear surrogate approximation. When K is moderate, exhaustive evaluation of all 2^K inclusion patterns is infeasible. I therefore approximate the log-odds function using a linear surrogate:

$$g(v) \approx b + \sum_{\ell=1}^{K} w_\ell v_\ell.$$

The coefficients w_ℓ are estimated by querying the model on a random sample of inclusion vectors $v^{(1)}, \ldots, v^{(n)}$ drawn uniformly from $\{0,1\}^K$. In the empirical implementation, $n = 64$ ablations per measure and $K \leq 30$ retrieved passages. Each evaluation yields an observation $g(v^{(i)})$, and the estimated coefficients $\hat{w}_\ell$ approximate average marginal log-odds effects:

$$\hat{w}_\ell \approx \mathbb{E}_{v_{-\ell}} \left[g(v_\ell = 1, v_{-\ell}) - g(v_\ell = 0, v_{-\ell}) \right].$$

Interpretation. Positive coefficients indicate passages that substantively support the generated motivation; coefficients near zero indicate irrelevance; negative coefficients indicate passages whose inclusion reduces regeneration probability.

Empirically, dependence patterns are sparse: only a small subset of passages typically receives substantively large positive weights. This sparsity is consistent with the interpretation that motivations are anchored in specific documentary statements rather than generated from diffuse contextual associations.

Illustration: VAT reform. Table 2 reports estimated passage-level contributions for the 1977 French VAT reform. For each generated motivation, positive weights are concentrated in passages describing the government's stated objectives, confirming that the summariser relies on policy-specific text.

Table 2: Attribution weights and placebo-test results across top ten retrieved passages

	Retrieved segments (1–10)									
	1	2	3	4	5	6	7	8	9	10
M_1	41.56	20.92	7.33	-0.24	-1.76	8.36	2.78	-0.69	-4.93	4.47
M_2	18.82	-4.13	15.29	-4.56	0.25	1.20	1.93	-3.47	-0.51	-3.26
M_3	2.88	8.99	1.19	0.76	-1.26	1.04	1.05	-1.40	-0.46	2.67

Note: Each entry reports the estimated marginal log-odds contribution of the corresponding passage to regenerating the motivation. Shaded cells indicate passages that pass the placebo-based grounding test at the 5% level. These entries exhibit positive marginal contributions that are statistically larger than would arise under random assignment of placebo labels. Further implementation details are provided in Section 3.2.2.

3.2.2 Placebo-Based Validation

Marginal contributions measure dependence, but a large positive coefficient does not by itself guarantee substantive relevance. A passage may receive a high weight because its linguistic structure resembles generic patterns learned during pretraining. To distinguish substantive grounding from incidental sensitivity, I implement a placebo-based diagnostic.

Real and placebo passages. For each measure, retrieved passages are partitioned into: (i) the first J segments, which contain policy-relevant material, and (ii) remaining segments drawn from the same document but semantically unrelated to the measure. Let $w_0, \ldots, w_{K-1}$ denote the estimated marginal coefficients.

Positive-only comparison. Because only positive coefficients indicate support, the test compares positive coefficients among real passages to the empirical distribution of positive coefficients among placebo passages:

$$\mathcal{P} = \{w_k : k \geq J,\ w_k > 0\}.$$

Monte Carlo inference. For each real passage with $w_j > 0$, define the p-value

$$p_j = \Pr(\tilde{w} \geq w_j), \qquad \tilde{w} \sim \mathcal{P},$$

estimated via Monte Carlo sampling. Small p_j indicates that w_j is unusually large relative to placebo passages and therefore reflects substantive grounding.

Interpretation. A generated motivation is deemed grounded when at least one associated passage exhibits a statistically significant positive contribution. This diagnostic provides a transparent map of documentary dependence and allows the researcher to focus manual verification on a small subset of supported passages. Appendix C.2 reports a complementary global test confirming that the observed grounding patterns are unlikely to arise by chance.

4 VAT in France

4.1 French Tax Data

After presenting the methodology, I now turn to its application to a substantive case: value-added tax (VAT) policy in France. Implementing this analysis requires the construction of a comprehensive dataset of French tax policy changes between 1960 and 2024. The dataset combines detailed fiscal information from official budget documents with contemporaneous legislative texts that record policy-makers' stated motivations. This dual structure allows both the quantitative identification of policy measures and the qualitative classification of their underlying rationales. The remainder of this section describes the construction of the dataset in two steps: first, the identification of tax policy changes in the *Voies et Moyens Tome I* (VMT), and second, the compilation of the textual corpus used to recover their motivations.

4.1.1 Identifying Tax Policy Changes

Primary Source: *Voies et Moyens Tome I* (VMT). To construct a comprehensive dataset of tax policy changes in France between 1960 and 2024, I systematically processed the information contained in each year's *Voies et Moyens Tome I* (VMT), an annex to the annual draft budget law (*Projet de Loi de Finances*). While most recent volumes of the VMT are accessible online, earlier

editions were not digitized. I therefore retrieved and scanned the missing volumes[3] from archival collections held by public institutions.

The VMT provides a detailed breakdown of the government's revenue projections and is published annualy. It covers both fiscal and non-fiscal revenues, showing how these are expected to evolve in the coming fiscal year. The VMT distinguishes between the mechanical evolution of tax bases, the effects of newly adopted legislation, and the delayed consequences of past reforms. Importantly, the section on the impact of past reforms also includes certain tax changes legislated between two VMTs, capturing some of the measures enacted outside the scope of the annual budget planning process. As some tax changes remain unaddressed, I will need to review relevant decrees ('décrets') and amended finance laws ('lois de finances rectificatives'). This will be incorporated into a next version of the paper.

Decomposition of Annual Revenue Changes. The change in net tax revenues from one year to the next, as presented in the VMT, is broken down into four main components[4]:

$$\begin{aligned}\Delta\text{Revenues} = \text{Spontaneous change} + \text{Scope and transfer measures} \\ + \text{Previous measures} + \text{New measures}\end{aligned} \tag{20}$$

1. **Spontaneous change**: This represents the natural (or endogenous) evolution of tax revenues at constant legislation. It captures changes due to economic dynamics, such as GDP growth, household income, consumption, inflation, and tax base expansion. It also includes the effects of tax progressivity, e.g., an increase in wages moving taxpayers into higher brackets.

2. **Previous measures**[5]: These are the effects of measures adopted in previous years, which continue to affect revenues through delayed application, phased implementation, or full-year effects. For example, the extension of a tax credit that starts mid-year in $t-1$ and applies fully in year t.

3. **New measures**[6]: These are new tax measures proposed in the current year's draft budget law (PLF). They include newly introduced taxes, rate changes, deductions, or credits that introduce a change in legislation.

4. **Scope and transfer measures**[7]: These reflect technical adjustments in how revenues are accounted for, such as reclassification of taxes from one budget entity to another, or rebudgetisation of previously off-budget revenues. A change in this category should not have any direct macroeconomic impact and will be excluded of the main dataset.

This structure allows for the identification of discretionary policy actions with potential macroeconomic implications—specifically, those classified as new measures (3), introduced in the current draft budget law, and previous measures (2), carried over from earlier reforms whose effects extend into the current fiscal year. In this paper, I will restrict the analysis of the dataset to new measures; integrating previous measures awaits reliable motivation mapping and is left to future work at this stage.

[3]The online VMT documents were unavailable for the years 1960–1983, as well as for 1988 and 1989.

[4]The explanations related to this breakdown is present in any year of the VMT documents. The years before 2000 do not include the component "Scope and transfer measures".

[5]Before 2009, this section is named "Other factors of variation".

[6]Before 2009, this section is named "Adjustment of rights".

[7]Before 2009, there is no dedicated section for this. Scope and transfer measures are present in the two former section (2) and (3).

Tax categories. The VMT systematically disaggregates annual revenue changes across a standardized set of 6 major tax categories[8]. This structured breakdown reflects the legal and economic distinctions between different revenue sources. The consistency of this classification across years ensures comparability and aids in constructing a coherent dataset of tax reforms[9].

I report the decomposition of revenues[10] for each of the major tax groups presented in the VMT in table 3 as well as some statistics about the data. For example, in 2024, VAT accounts for 38% of total gross revenues and 6% of GDP. Between 1980 and 2024, the VMT records a total of 444 changes in VAT[11].

Table 3: Gross Tax Revenue Breakdown by Source (France, 2024)

Tax Category	**Share of Revenue (%)**	**As % of GDP**	**#New Measures**	**#Old Measures**
Personal Income Tax	24.3%	3.89%	359	1940
Other Direct Taxes	7.2%	1.15%	185	284
Corporate Income Tax	18.5%	2.96%	137	1459
Energy Consumption Tax	3.8%	0.60%	65	114
Value-Added Tax (VAT)	37.8%	6.06%	140	324
Other Indirect Taxes	8.5%	1.36%	243	295
Gross Tax Revenues	**100.0%**	**16.02%**	1145	4444

Dates. The data collected is at the yearly frequency. To convert the dataset to the quarterly frequency, the following procedure is applied.

Measures that have been enacted during the annual budget draft are enacted the 1st of January of the coming year. This represent the majority of the measures. However, there are exception: some measures are applied by anticipation, some others are planned to be applied in the coming years. This information is partly taken from the data provided by the *Institut des politiques publiques* that collects the date of implementation of the different taxes[12].

Measures coming from Law finances that are legislated in between the annual budget procedure are allocated to the quarter where it has been legislated. The same applies to other type of laws that do not correspond to LF or LFR but still affect the ressources of the state.

[8] I merge the two categories related to "Other direct taxes" in one single categories for simplicity.

[9] Categories are broadly consistent over the years appart from few example that will be documented in the Appendix.

[10] The higher figures for the tax-to-GDP ratio reported by institutions like the OECD and IMF arises from differences in definitions and the scope of what is considered "tax revenue." They usually include compulsory social security contributions in their tax revenue calculations, which is not the case in the figures presented in the annual draft budget law. In France, these contributions are substantial and significantly impact the tax-to-GDP ratio reported.

[11] This number excludes the new measures related to "Scope and transfer" when identifiable.

[12] We timestamp measures using the *Institut des politiques publiques* (IPP) statute-level database, which records for each change the legal basis (law/decree/arrêté or social-partner agreement), the publication date in the Official Journal (JORF), the effective date, and links to the source on legifrance.gouv.fr. For tax schedules, coverage is historical (income tax back to 1914; social contributions to 1945; most items from the 1980s) and maintained with the OpenFisca community. In my quarterly mapping, I use IPP effective dates; when unspecified, I fall back to the JORF publication quarter.

4.1.2 Corpus of Text to Identify the Motivations behind the Tax Changes

A final step in building the dataset is to curate a corpus of contemporaneous texts which likely include the motivations behind the different granular tax changes. As highlighted by Romer and Romer, the sources should be carefully selected to ensure they reflect the political, economic, and social context of the time when the tax changes were proposed and enacted. In addition, the source should be reliable and consistent over time. To achieve this, I collect the Senate reports as the primary source for analysis from 1960 to 2024.

In the analysis of finance bills, Senate reports provide a particularly useful entry point. They are often more critical and technical than those of the National Assembly, making them well suited for identifying the government's underlying motivations behind changes in taxation. By highlighting inconsistencies, emphasizing the fiscal consequences of certain choices, or questioning the official justifications advanced by the government, the Senate offers insights that go beyond the surface of the legislative text. In addition, because the Senate examines the text after the Assembly's first reading, its reports also indicate which measures were adopted, amended, or deleted upstream. My current analysis therefore relies on Senate reports both to trace the motivations behind tax reforms and to verify which provisions survived the Assembly's first examination.

That said, the analysis of reports from the National Assembly would have their own advantages. They tend to align more closely with the government majority, and thus provide the most direct view of the legislative choices of the ruling coalition. While they are less critical in tone, they are essential for confirming the government's political priorities and for tracking in real time the measures approved at each stage. For now, my focus remains on the Senate's perspective, which combines a critical reading of motivations with information on legislative outcomes, but adding the Assembly reports could be a natural extension of the corpus.

4.2 Applying SANI

4.2.1 Embedding model and LLM choice

To implement the SANI framework, I first select a pretrained embedding model to represent the textual content of policy documents in vector form. I use the `text-embedding-large` model developed by OpenAI, which provides a one-to-one mapping between each text segment and a high-dimensional embedding vector.

For large language model (LLM) operations, I employ a two-model strategy that balances performance and reproducibility. The primary outputs of the SANI procedure—summarization of retrieved segments and classification of policy motivations—are generated using `gemini-3.0-pro`. The use of a very large model at these stages offers high performance in natural language understanding and reasoning, ensuring high-quality generation of structured summaries and consistent application of categorization rules.

For robustness and interpretability analyses described in Section 3, I rely on the open-weight model `Qwen-8B`. Using an openly available model allows full reproducibility of the marginal narrative contribution analysis. The combination of these two models thus leverages the superior generative capabilities of frontier proprietary systems for the main classification outputs, while preserving the replicability and auditability afforded by open-weight alternatives for robustness checks. That said,

the entire SANI procedure can, if needed, be implemented exclusively with open-weight models without altering its logic or structure.

This separation between generation and marginal source contribution stages is deliberate and methodologically valid. The computation of source contributions operates conditional on a fixed output: its goal is to quantify how individual retrieved segments influence that output, regardless of how it was produced. Consequently, the interpretability analysis can be conducted with an alternative or open-weight model without affecting the validity of the results. In principle, the same procedure could be applied to outputs generated by any model or even by human analysts, provided the classification or summary is held constant.

4.2.2 Retrieval

The first step in applying SANI to French VAT policy is to retrieve, for each identified tax change, the passages in contemporaneous Senate reports that contain the government's stated motivations. Because retrieval quality directly affects all subsequent stages, this subsection evaluates the performance of the semantic retriever and documents the trade-offs involved in choosing segmentation length L and the number of text segments retrieved K. Using a manually constructed benchmark, I assess both the optimal parameter configuration and the value of exploiting document structure relative to a fully unsupervised segmentation approach.

Benchmark design: golden articles. Each VAT measure in the *Voies et Moyens Tome I* (VMT) is manually linked to the specific article of the Senate report that discusses the corresponding provision of the budget law. I refer to this as the *gold article*. Because French budget laws are highly structured—each article of the draft law introducing a specific tax measure, followed by a dedicated Senate discussion—the mapping between policy events and relevant text is clear. However, in the retrieval stage, documents are divided into fixed-length text segments of length L, and a single gold article spans several segments. The goal is therefore not to retrieve one exact chunk, but to ensure that among the top-K retrieved segments there is sufficient overlap with the true gold article.

Ensuring coverage. Performance is assessed by verifying, for each event, whether at least one of the top-K retrieved segments belongs to the corresponding gold article. This criterion captures whether the model successfully localizes the relevant discussion. Across segment lengths $L \in \{512, 1024, 2048, 4096\}$, results are highly robust: in nearly all cases, at least one segment from the gold article appears among the top ten retrieved passages, and on average 5.5 of the ten retrieved segments overlap with the gold article.[13] Increasing K or L beyond these ranges yields only marginal improvements, indicating that retrieval quality is high even with modest parameter values. The configuration $L = 1024$ is adopted as the baseline, but given the low computational cost of the full procedure, alternative values of L are also implemented and will be reported as robustness checks in a future version of the paper.

Supervised versus unsupervised segmentation. A key practical question concerns whether retrieval requires explicit alignment with the legal structure of the documents. In the *supervised* approach, the Senate reports are preprocessed so that segment boundaries coincide with article boundaries. In the *unsupervised* approach, the text is divided mechanically into equal-length segments

[13] For $L = 2048$, this is true for 97% of the measures.

without any prior formatting or alignment. Both methods deliver nearly identical recall among the top ten retrieved passages: the supervised variant attains full coverage with only a slightly smaller K for segment lengths $L \in \{512, 1024\}$. Given that the supervised approach entails substantial additional work, parsing, aligning, and annotating article boundariesk the improvement in accuracy is minor relative to its cost. This result is central to the contribution of SANI: it suggests that reliable narrative retrieval can be conducted directly on raw legislative text, without the need for costly manual preparation of sources. By removing this preprocessing requirement, the procedure markedly lowers the barriers to scaling the narrative approach across countries and policy domains.

Overall, the retrieval stage consistently identifies the correct Senate passages explaining each VAT measure. The method achieves near-perfect recall at low computational cost and without relying on document-specific alignment. This ensures that the subsequent summarization step operates on the relevant textual evidence for each policy change, while preserving the scalability that motivates the use of AI-assisted narrative identification.

4.2.3 Summarization

Having established that retrieval reliably identifies the relevant passages for each VAT measure, the next step condenses these passages into structured narratives that summarize the government's stated motivations. This stage applies the summarization operator $\mathcal{S}$, which transforms the retrieved text segments $\hat{R}_{j,t}$ into a compact and standardized representation $\hat{S}_{j,t}$. The goal is to translate lengthy legislative discussions into concise, factual statements that preserve the logic and evidential grounding of the original text.

Prompt design. Summarization is implemented through a LLM instructed to extract the government's stated motivations from the retrieved passages and to present them in a structured format. The model receives a fixed prompt of the following form:[14]

English translation of the prompt:

Based on the [EXCERPTS] from the Senate report on the draft budget law for {year}, what are the main motivations of the government related to the tax change {tax type}, description: {description}? Summarize these motivations and limit yourself to a maximum of three main ones.

- Rely only on the content of the excerpts, without adding interpretation.
- Consider only motivations directly linked to the described measure.

[EXCERPTS]
[Excerpt no. 1] ... [End Excerpt no. 1]
[Excerpt no. 2] ... [End Excerpt no. 2]
[...]

Prompt 1: Summarization Prompt

[14]The prompt specifies as well a uniform output structure across all measures: motivations must appear as bullet points beginning with "–", and listed directly without introductory text.

Dynamic fields. The placeholders {`year`}, {`tax type`}, and {`description`} in the prompt are automatically populated with information specific to each VAT measure. The variable `year` indicates the fiscal year of the corresponding budget law, `section` identifies the legal article or provision introducing the tax change, and `description` provides a short textual description of the measure directly drawn from the French budgetary documents *Voies et Moyens Tome I*.

Fidelity and grounding. The credibility of the summarisation stage hinges on the extent to which generated motivations remain faithful to the retrieved passages. I assess this using two complementary approaches. First, grounding is evaluated through the perturbation-based attribution test introduced in Section 3, which measures how the probability of regenerating a given motivation responds to ablating individual retrieved segments. For 98.5 percent of generated motivations, at least one retrieved passage exhibits a significant positive attribution weight at the 2.5 percent level, indicating that the model's output depends on the supplied evidence rather than on prior knowledge. Together, these results show that $\mathcal{S}$ functions as a faithful compression operator that extracts motivations without distortion or hallucination. Second, manual inspection of a benchmark sample of VAT measures is underway to provide additional qualitative validation.

The structured narratives produced by this step serve as direct inputs for the classification operator $\mathcal{C}$, which maps each summarized motivation into a policy category according to codified decision rules. The next subsection applies this operator to distinguish exogenous from endogenous VAT changes based on their stated motivations.

4.2.4 Design of Classification Categories

Whereas the retrieval and summarization stages of SANI are general, the design of classification categories is inherently application-specific. It defines the conceptual boundary between types of policy actions and therefore determines the interpretation of the resulting narrative shocks. In this application, the objective is to classify value-added tax (VAT) measures as *endogenous* or *exogenous*, following the narrative taxonomy developed by Cloyne (2013) for the United Kingdom. This framework provides economic criteria for distinguishing tax changes responding to contemporaneous macroeconomic conditions from those motivated by long-run structural, ideological, or institutional considerations.

Table 1 recall the categories defined by Cloyne (2013), along with representative examples. In this taxonomy, *endogenous* measures are those explicitly linked to current economic or fiscal conditions—for instance, to stimulate demand, stabilize activity, or address short-term budget pressures. By contrast, *exogenous* measures are justified by objectives unrelated to the current state of the economy, such as improving efficiency, ensuring long-run fiscal sustainability, or advancing ideological or institutional reforms. This distinction is conceptually consistent with Romer and Romer (2010), who classify U.S. tax changes according to their stated motives, but adapted by Cloyne to more granular categories.

Operationalization in SANI. In practice, the classification operator $\mathcal{C}$ applies codified decision rules to the structured motivations $\hat{S}_{j,t}$ generated by the summarization stage. Each subcategory is assessed independently using a dedicated prompt, yielding binary outcomes that indicate whether a given motivation supports that category. A single prompt example, corresponding to the *external* category, is reported below; the remaining category prompts are presented in Appendix D.2.

English translation of the classification prompt (External / EXT):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "External (EXT)"?

Definition of External (EXT):
A passage is classified as External if the measure is motivated by external economic pressures, trade obligations, or international agreements.

This includes:

- Measures adopted in response to balance-of-payments problems, exchange rate pressures, or the need to protect reserves.
- Tax modifications explicitly linked to supporting exports, foreign trade operations, or competitiveness in international markets.
- Adjustments mandated by commitments to international organizations or treaties (e.g., WTO, EU, EFTA).
- Policy changes justified by concerns about investor confidence or market reactions to the government's financing needs.

*Do **not** classify as External if:*

- The motivation is purely domestic (e.g., deficit reduction, consolidation, demand management, social equity, environment).
- The change is a routine adjustment unrelated to trade, finance, or international agreements.
- The measure concerns only domestic consumption, revenue, or distribution without an external framing.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 2: Classification Prompt for the "External (EXT)" Category

Group	Sub category	Explanation and examples
Endogenous (N)	1. Demand management (DM)	• Targeting the aggregate level of demand e.g. to boost investment, consumption, growth, or curb inflation. • Specific help to households and individuals by stimulating disposable income. • Dealing with a balance of payments crisis via demand.
	2. Supply stimulus (SS)	• Certain help for businesses during a downturn (e.g. NIC cut). • Short term sector support (e.g. targeted tax cuts for a sector).
	3. Deficit reduction/BoP crisis (DR)	• Direct measures to deal with a budget or external deficit contemporaneously caused.
	4. Spending-driven (SD)	• Taxes which fund specific spending commitments.
Exogenous (X)	1. LR performance (LR)	• Measures to improve competitiveness, productivity, efficiency and long-run growth (but not taken to offset a shock). • Simplification and deregulation measures. • Long-term support for business or sectors of the economy.
	2. Ideological (IL)	• Long-term social or political goals, independent of their effect on performance and not to offset current shocks. • Some anti-avoidance measures (where no other motive is given).
	3. External (ET)	• Court rulings and enforcement of directives.
	4. Deficit consolidation (DC)	• Measures to lower inherited deficit for reasons of economic philosophy or to offset current actions in the future. • Does not include actions forced on the government, or decisions contemporaneous motivated by a current shock.

Figure 1: Classification of Tax Measures Following Cloyne (2013)

4.3 Constructing the VAT shock

4.3.1 Preliminary Narrative Shock

To construct the exogenous share for each VAT measure, I apply the marginal narrative contribution framework of Section 3 to the calibrated classification scores obtained for every summarized motivation. In principle, the exogenous share would combine not only the singleton effects of individual motivations but also their interaction effects when presented jointly. However, preliminary exploration indicates that higher-order interactions tend to blur the motivation-specific signal in ways that are not yet well understood and are the subject of ongoing investigation. I therefore rely solely on singleton marginal effects: for each motivation, the category with the highest calibrated score is selected via a simple max rule, and measure-level exogenous support is computed by aggregating these assignments across motivations. The resulting exogenous shares display substantial heterogeneity across VAT measures, ranging from fully endogenous to fully exogenous cases, with a broad distribution in between. Figure 2 documents this empirical distribution for the universe of VAT changes.

Figure 3 plots the resulting narrative-weighted exogenous VAT shock series over the full sample, alongside the corresponding series of all legislated VAT changes. The exogenous series still displays pronounced spikes, while remaining comparatively muted during periods in which VAT changes were motivated primarily by short-run fiscal or stabilisation concerns. The 1968 shock is muted because its stated motives of revenue compensation, financing deduction expansions, and sectoral alignment were fully endogenous.

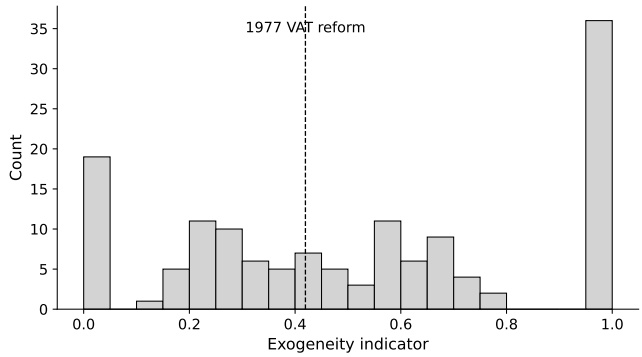

Figure 2: Distribution of Exogeneity Shares Across VAT Measures

Note: The histogram reports the distribution of exogeneity shares constructed from singleton marginal narrative contributions. Mass at zero corresponds to measures whose motivations are entirely endogenous, while the spike at one reflects measures classified as fully exogenous. Intermediate values indicate mixed motives and partial exogeneity. Shares are based on the max-assignment rule described in the text.

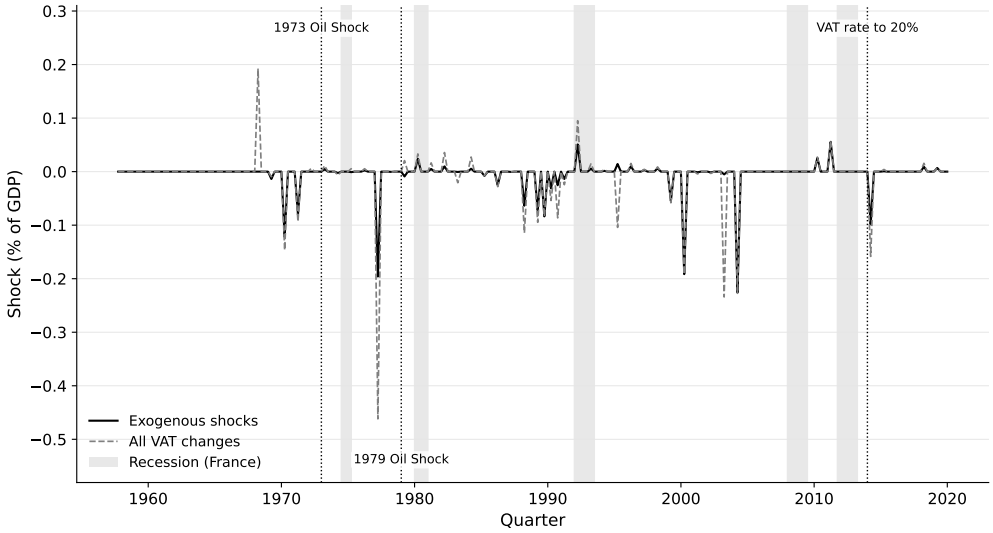

Figure 3: Time Series of Narrative-Weighted Exogenous VAT Shocks and All VAT Changes

Note: The figure plots the time series of narrative-weighted exogenous VAT shocks together with the corresponding series of all VAT changes. Grey shaded areas denote recession periods in France, while vertical dotted lines mark major macroeconomic and fiscal events. Shocks are expressed as a percentage of GDP.

4.3.2 A text-augmented VAT policy rule

Beyond delivering narrative shocks, the SANI outputs also permit a parametric characterisation of systematic VAT behaviour. By treating the calibrated motivation intensities as observable components of the policymaker's information set, I estimate a text-augmented policy rule that complements the narrative identification. Following the policy-rule framework introduced in Section 1, and in the spirit of the text-augmented approach of Aruoba and Drechsel (2025), the unobserved information set is approximated using both macro-fiscal variables and the qualitative motives extracted by SANI. This allows systematic VAT behaviour to be modelled jointly from quantitative fundamentals and stated

motivations, while SANI continues to isolate the unsystematic component through narrative evidence.

To operationalise this policy rule, I first introduce the relevant observable covariates. Let X_t denote a vector of macro-fiscal controls (output gap, inflation, deficit, and lagged VAT variables). For each VAT measure (j,t) and each classification category $k \in \{1,\dots,K\}$, the randomised source-dependence procedure produces marginal source contributions $\{\hat{w}^{(k)}_{j,t,\ell}\}_{\ell=1}^{L_{j,t}}$, where $\hat{w}^{(k)}_{j,t,\ell}$ measures how strongly motivation $M_{j,t,\ell}$ shifts the log-odds of assignment to category k.[15]

Direction-adjusted narrative intensities. The marginal source contributions lie in $[0,1]$ and measure the strength of narrative support for category k, independently of whether the VAT measure raises or reduces the tax burden. In the policy-rule regression, however, the effect of a motivation must reflect the economic direction of the measure. I therefore construct the category–specific intensity as

$$m^{(k)}_{j,t} = \operatorname{sign}(\Delta\tau_{j,t}) \sum_{\ell=1}^{L_{j,t}} \hat{w}^{(k)}_{j,t,\ell}, \qquad k = 1,\dots,K. \tag{21}$$

Thus, the magnitude of $m^{(k)}_{j,t}$ increases with the cumulative narrative support for category k, while its sign reflects whether the measure raises or lowers the VAT burden.

Collecting these into $m_{j,t} = (m^{(1)}_{j,t},\dots,m^{(K)}_{j,t})'$, I estimate the following text-augmented VAT policy rule:

$$\Delta\tau_{j,t} = \alpha + \beta' X_{t-1} + \gamma' m_{j,t} + \delta' Z_t + u_{j,t}, \tag{22}$$

where β captures systematic responses to lagged macro-fiscal conditions, γ loads the direction-adjusted narrative intensities, and $u_{j,t}$ is the innovation. The vector Z_t collects contemporaneous changes in other tax instruments, such as personal and corporate income taxes, constructed from the same dataset. Including Z_t filters out systematic comovement across fiscal instruments and isolates the component of VAT policy uniquely associated with its own narratives. The fitted values represent the systematic component of VAT policy conditional on $(X_{t-1}, m_{j,t}, Z_t)$, and the residual $u_{j,t}$ is interpreted as the VAT policy shock orthogonal to both quantitative and narrative information.

In the current version of the paper, I do not yet report estimates of (22); the empirical analysis instead relies on the baseline source-augmented narrative shocks.

4.4 Macroeconomic effects of exogenous VAT shocks

Interpretation of the empirical estimates. The empirical analysis in this section adopts a benchmark implementation that is nested within, but stronger than, the moment-based identification framework developed in Section 1.

I construct an aggregate narrative-weighted tax measure defined as $\Delta\tau_t = \sum_j p_{j,t}\,\Delta\tau_{j,t}$ and treat this object as orthogonal to contemporaneous structural disturbances.

Formally, the maintained restriction is

$$\mathbb{E}(p_{j,t}\,\Delta\tau_{j,t}\,\varepsilon_t) = 0,$$

which is stronger than the graded moment restrictions derived in Section 1.3. Under this assumption,

[15] See the subsection on the randomised source-dependence method for details of the log-odds surrogate model and the definition of marginal source contributions.

narrative support acts as a proxy for the latent orthogonal component of policy actions, so that the resulting estimates recover the causal effect of orthogonal tax variation insofar as residual narrative misclassification is mean-independent of structural shocks.

This implementation imposes full orthogonality of the narrative-weighted policy measure rather than exploiting variation in $p_{j,t}$ to identify the orthogonal-class moment through cross-moment dispersion. It therefore delivers an empirical object directly comparable to traditional narrative shock series and facilitates comparison with existing estimates in the literature.

The graded framework developed in Section 1, however, allows identification under weaker restrictions by leveraging systematic variation in narrative support, without requiring that $p_{j,t}\Delta\tau_{j,t}$ itself be orthogonal. Implementing estimators that fully exploit this variation is the natural next step and does not alter the interpretation of the benchmark results reported here.

Econometric specification. To estimate the dynamic effects of fiscal shocks, I rely on a local projection framework in the spirit of Jordà (2005). Specifically, for each horizon h, I estimate

$$\Delta^h y_{t+h} = \alpha_h + \beta_h \Delta\tau_t + \Gamma_h X_t + u_{t+h}, \tag{23}$$

where $\Delta^h y_{t+h}$ denotes the h-period-ahead change in the outcome of interest, $\Delta\tau_t$ is the narrative-based tax shock, and X_t is a vector of controls including six lags of real GDP growth and inflation. The coefficients β_h trace the impulse response of the outcome variable to the tax shock. Standard errors are computed using Newey–West corrections to account for serial correlation across horizons.

Data and Detrending. All data series are obtained from the French national accounts published by INSEE. We use quarterly data on GDP, its expenditure components, and the GDP deflator, ensuring internal consistency across the usual national accounting identity. All nominal aggregates—including GDP and its subcomponents—are deflated by the GDP deflator to obtain real series. We then remove long-term trends following the procedure of Gordon and Krenn (2010): (i) we regress real GDP on a quadratic time trend, and (ii) divide all real variables by the fitted quadratic trend. This transformation expresses each component as a percentage of trend GDP, thereby preserving the exact decomposition of GDP into its constituent parts. Compared to logarithmic detrending, the Gordon–Krenn approach has the advantage of maintaining the accounting identity across the disaggregated components of GDP. In

4.4.1 Empirical results

Baseline specification. Figure 4 presents the estimated responses of real GDP, inflation, VAT revenues, and the government deficit to a VAT increase of one percent of GDP. Real GDP declines persistently, reaching a trough of about three percent below trend around the twelfth quarter, before recovering gradually and returning to zero by roughly the twenty-second quarter. Inflation exhibits a short-lived spike: it rises sharply on impact, peaking around the fourth quarter as the higher indirect tax is passed through to consumer prices, and then declines steadily toward zero.

VAT revenues increase mechanically on impact, and the deficit improves for the same reason, before deteriorating once the fall in GDP depresses the broader tax base. As activity normalizes, the deficit gradually returns toward balance.

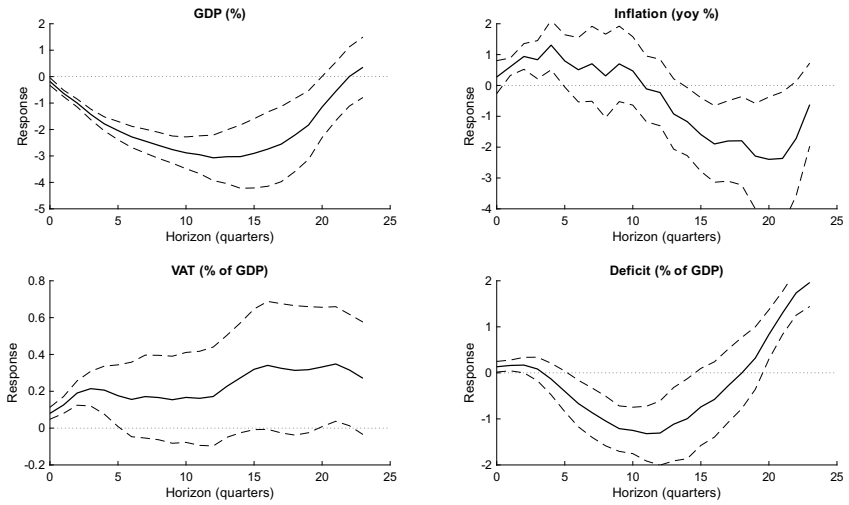

Figure 4: Responses to a +1%-of-GDP shock in VAT (68% Confidence Bands)

Overall, the results indicate that a VAT increase generates a moderate but prolonged contraction in output and a brief inflationary surge. The magnitude of the GDP response is strikingly similar to that reported by Romer and Romer (2010) for broad-based tax increases, which also produce output losses of around three percent.However, VAT is generally regarded as less distortive than the income and profit taxes that dominate the Romer–Romer measures, so one might have expected a weaker output response.

The fact that both tax changes produce effects of comparable size might be driven by differences in the monetary-policy environment surrounding the two shocks. Broad-based tax increases in Romer and Romer (2010) could plausibly trigger monetary accommodation that mitigates the short-run contraction, whereas a VAT hike generates a short-lived but pronounced spike in inflation, which may limit the scope for such accommodation. While this interpretation is only suggestive, it offers a consistent explanation for why a conceptually less distortive tax change produces output effects similar to those observed following broader tax hikes.

5 External Validation: The Romer and Romer (2010) Benchmark

This section externally validates the proposed framework using the canonical narrative tax series of Romer and Romer (2010). The Romer–Romer benchmark is a natural external validation target: it is widely used in empirical macroeconomics, it is grounded in close readings of contemporaneous policy documents, and it has well-established macroeconomic implications.

The validation proceeds in two steps. I first apply the probability-based procedure to edited Romer–Romer narratives. Rather than asking the model to reproduce a binary label directly, I use the RR narratives to generate structured motivations, assign exogeneity probabilities at the motivation level, and aggregate them into a policy-level probability. This benchmark isolates the probability-construction stage of the framework from document retrieval while preserving the original RR narrative evidence. I then build a new dataset of routine tax changes in the U.S. and apply the full procedure to U.S. policy documents, thereby relaxing the assumption that curated narratives are available ex ante.

5.1 Validation using the Romer–Romer narratives

This subsection evaluates whether the probability procedure developed in Section 2 recovers the economic content of Romer and Romer (2010) when it is applied to the edited narratives used in the original study. The goal is not to reinterpret the historical record or to force an automated binary relabeling of the RR episodes. Instead, the exercise asks whether the procedure introduced in the paper assigns systematically higher exogeneity support to RR-exogenous episodes and whether the resulting probability-based shock series reproduces the benchmark macroeconomic response.

By conditioning on the curated Romer–Romer narratives, this benchmark abstracts from document retrieval while still preserving the summarization, motivation decomposition, and probability aggregation stages of the framework. It therefore provides a test of whether the probability object constructed by the methodology carries a similar identifying content as the original RR narratives.

5.1.1 Setup

The Romer–Romer replication material documents 50 major federal tax changes in the United States between 1945 and 2007. For each episode, the authors provide a detailed narrative discussion of the motivations underlying the reform, drawing on contemporaneous sources such as presidential speeches, budget documents, and Congressional records. Each narrative concludes with a classification of the tax change as either exogenous or endogenous.

To construct the benchmark validation task, I extract the narrative text describing each policy episode and remove the portion in which the authors explicitly state and justify their classification. In particular, I delete the sentence reporting the final classification along with the two immediately preceding sentences, which often contain direct cues revealing the intended label (see Appendix E.1.1 for examples). The remaining narrative text is used as input to the language model.

5.1.2 From edited narratives to policy-level probabilities

The edited RR narrative does not enter the benchmark as a single paragraph to be mapped directly into a binary label. Instead, each edited narrative is first summarized into a set of stated motivations. Because a single RR episode can contain several rationales, the same policy may generate multiple motivations, each of which is scored separately. These motivation-level outputs are then mapped into exogeneity probabilities and aggregated into a policy-level structural probability.

The construction follows the cross-implementation aggregation described in Section 3; see in particular equation (18). Throughout this benchmark, the object of interest is therefore the policy-level narrative support measure $\bar{p}_{j,t}$, rather than any single implementation-specific estimate $\hat{p}_{j,t}^{(r)}$.

5.1.3 Probability validation against RR classifications

Table 4 reports the original Romer–Romer binary label and asks whether RR-exogenous episodes receive systematically higher policy-level probabilities. The mean probability rises from 0.42 for RR-endogenous episodes to 0.58 for RR-exogenous episodes, and the share above tighter thresholds also rises.

Table 4: Policy-Level Structural Probabilities by Original RR Classification

RR class	**N**	**Mean** `p_mean_models`	**Share** $p \geq 0.5$	**Share** $p \geq 0.7$
RR endogenous	24	0.42	33.3%	4.2%
RR exogenous	26	0.58	73.1%	19.2%
Difference (exo - endo)	–	0.16	39.7 pp	15.1 pp

Note: Entries are computed from the 50 edited RR narratives. `p_mean_models` is the policy-level probability from the average-based stage-2 specification.

The separation is informative but far from complete, and that is precisely why a continuous object is useful. The relevant question is not whether the procedure reproduces a single RR label exactly, but whether episodes judged exogenous by Romer and Romer receive systematically more documentary support for exogeneity. Under a probability-based implementation, ambiguous cases are down-weighted rather than being forced into an all-or-nothing inclusion rule. Appendix E.2 provides training leakage checks.

5.1.4 Macroeconomic implications

I next assess whether these policy-level probabilities preserve the macroeconomic content of the Romer–Romer shock series. For GDP, I now match the published RR benchmark exactly: quarterly real GDP growth is regressed on the contemporaneous tax shock and twelve lags of that shock, together with eleven lags of GDP growth, over the 1950Q1–2007Q4 sample. The reported output response is the implied cumulative effect on GDP, and all curves are shown with 68% confidence bands. This subsection focuses on GDP because that is the benchmark specification reported in Romer and Romer (2010).

Figure 5 compares the original RR series to three estimators built from `p`. Panel (a) reports the hard-threshold estimator with $\bar{p} = 0.5$, which converts the continuous support measure back into a binary-style filter. Panel (b) reports the soft-weighted estimator, which uses the full ranking informa-

tion in the probabilities. Panel (c) reports the two-type estimator evaluated at $p = 1$, corresponding to the most exogenous endpoint implied by the affine mixture specification.

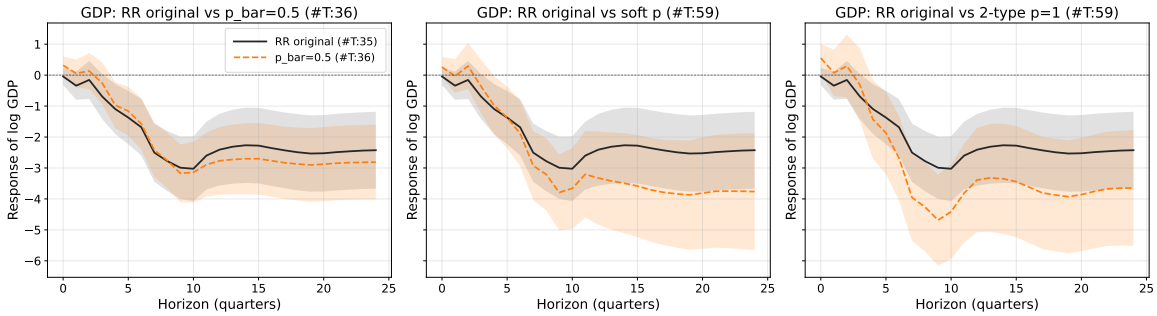

Figure 5: Romer–Romer GDP Responses: Original Series versus Probability-Based Estimators

Note: Each panel reports the exact Romer–Romer GDP specification. The black line is the original RR exogenous shock series. The colored line is the corresponding estimator constructed from `p`. Shaded areas denote 68% confidence bands.

The three panels are informative in different ways. The the original RR series is slightly attenuated relative to hard-threshold estimator at medium horizons, but at short horizons the benchmark contraction is well tracked. The black RR benchmark itself reaches a peak contraction of roughly 3 log points around quarter 10, in line with the original Romer–Romer result. The soft estimator and the two-type $p = 1$ endpoint are more contractionary from medium horizons onward, indicating that the full probability ranking retains economically relevant information that is discarded by a binary cutoff. At the same time, all three probability-based estimators remain much closer to the canonical RR response than the series obtained by treating all tax changes as equally exogenous; Appendix E.4 shows that the all-tax-changes series is markedly attenuated.

Figure 6 unpacks the two-type estimator more directly. The left panel reports the two endpoint responses, $p = 0$ and $p = 1$. The right panel then shows the implied linear mapping at horizon $h = 10$, which is where the exact-RR benchmark reaches its peak contraction in the current implementation. Read this way, the two-type estimator is tracing how the estimated response moves between the low-support and high-support endpoints as the policy-level exogeneity probability increases. In the RR benchmark, the gap between the two endpoints is already substantial by $h = 10$, with the high-p endpoint much closer to the canonical RR contraction than the low-p endpoint.

Taken together, these results show that policy-level exogeneity probabilities extracted from edited RR narratives recover the central macroeconomic content of the original benchmark. If moving toward higher-p tax changes systematically pulls the estimated response toward the RR benchmark and away from the attenuated all-tax-changes series, then the probability pipeline is capturing the same identifying content as the RR narratives themselves. The next subsection extends the validation by applying the full procedure to U.S. policy documents, thereby assessing whether similar conclusions obtain once retrieval and summarization are carried out end-to-end.

5.2 End-to-end extension using U.S. policy documents

The previous subsection validates the probability-construction stage while holding fixed curated Romer–Romer narratives. I now relax that curated-narrative assumption and apply the full procedure to

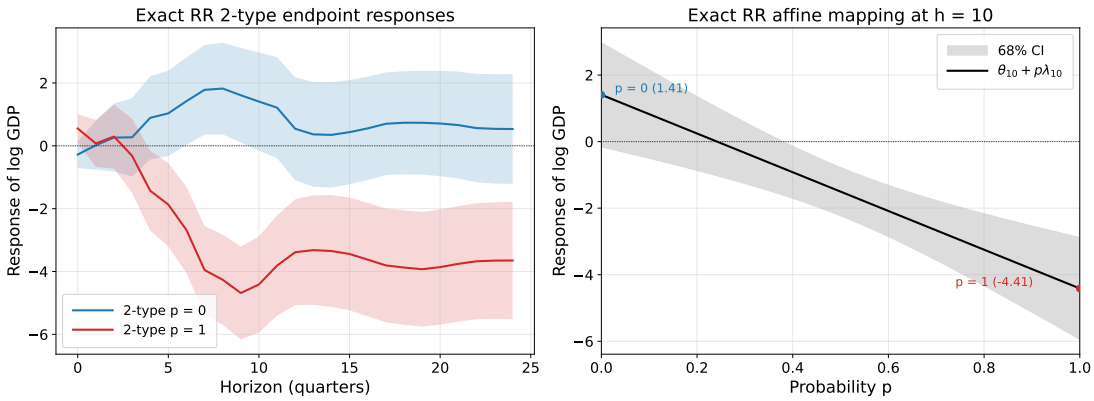

Figure 6: Two-Type RR Benchmark: Endpoint IRFs and the Mapping at $h = 10$

Note: The left panel reports the exact-RR two-type endpoint responses for $p = 0$ and $p = 1$, each with 68% confidence bands. The right panel plots the affine response mapping $\theta_h + p\lambda_h$ at horizon $h = 10$, together with its 68% confidence band.

routine U.S. federal tax measures using contemporaneous policy documents. The objective is to ask whether newly constructed budget-based shocks become more RR-like once documentary support for exogeneity is used to filter or weight tax changes.

5.2.1 U.S. tax data and coverage

Existing narrative coverage. The narrative tax dataset of Romer and Romer (2010) identifies major federal tax actions in the United States over the postwar period, focusing on large and salient legislative episodes. The series includes comprehensive tax reforms and major standalone acts, such as income tax reforms, Social Security legislation, and federal highway funding acts, that generate sizable and clearly identifiable changes in aggregate tax liabilities. By construction, it abstracts from routine fiscal adjustments and from smaller tax provisions embedded in annual budgetary processes. As a result, the RR series provides a measure of large, infrequent tax shocks, but it offers only partial coverage of legislated tax changes and at an aggregated tax-instrument level.

Extension. To recover the tax changes omitted, I construct a comprehensive dataset of legislated federal tax measures by systematically processing each year's *Budget of the United States Government* from 1950 to 2024. For each fiscal year, I extract all reported changes in federal taxation and record them at the highest level of detail available, including a disaggregation by tax instrument whenever possible. The same extraction procedure is applied to the major tax acts identified by Romer and Romer (2010), yielding a finer tax-level decomposition than that available in the extension of Mertens and Ravn (2013).

For the end-to-end validation below, I focus on the budget-based portion of this broader dataset. That choice is deliberate: the annual budget volumes provide a large set of routine tax measures that mostly fall outside the canonical RR benchmark and therefore furnish an out-of-sample environment in which to test whether the probability-based procedure recovers economically meaningful identifying content.

Table 5 reports the composition of the resulting budget dataset. It aggregates all tax changes

explicitly proposed in the yearly *Budget of the United States Government* over the sample period and classifies them into six major federal tax categories. The table therefore describes the structure of tax policy adjustments as they appear in executive budget proposals, not the universe of all federal tax legislation. In the final dataset, each tax change is associated with the quarter in which it becomes effective. In particular, when no specific quarter is mentioned, the tax change is assigned by default to the first quarter of the year.

Table 5: Distribution of New Tax Measures by Category (United States)

Tax Category	**# New Measures**
Individual Income Tax	371
Corporate Income Tax	361
Consumption Tax	238
Payroll / Social Insurance Tax	114
Other Receipts	89
Wealth Transfer Tax	17
Total	**1190**

5.2.2 Document corpus

A final step in constructing the U.S. tax dataset is to assemble a corpus of contemporaneous documents that records policymakers' stated motivations for legislated tax changes. Sources are selected to reflect the policy rationale articulated at the time of enactment and to remain consistent and comparable over long horizons.[16] The broader U.S. corpus relies on two recurring source families: the *Budget of the United States Government* and the *Annual Report of the Secretary of the Treasury*.

For the end-to-end benchmark in this subsection, the relevant document source is the annual Budget itself. These budget volumes are the full-document analogue of the edited RR narratives used above: instead of starting from hand-curated narrative excerpts. The implementation follows the same settings adopted in the French application; details are provided in Section 4.2.

5.2.3 Distribution of estimated exogeneity support

Figure 7 reports the distribution of the policy-level probability p across the budget measures that enter the quarterly shock construction. Unlike the earlier binary benchmark, the distribution is overwhelmingly interior. In the current matched sample, the mean and median are both about 0.54, 58% of measures have $p \geq 0.5$, and 22% have $p \geq 0.7$, while neither exact zero nor exact one occurs once multiple motivations are aggregated at the policy level.

This is precisely the environment in which a probability object is useful. Routine budget measures often combine several stated motivations, so the aggregation step naturally produces graded support rather than literal corner solutions. The right panel of Figure 7 makes that filtering margin transparent:

[16] Romer and Romer (2010) draw on a broad set of contemporaneous sources, including legislative debates and discussions in congressional chambers. Some of these materials are not systematically used here, primarily because their access and coverage are uneven over time. The analysis therefore focuses on recurring executive and Treasury documents that are consistently available throughout the sample period.

moving from all measures to high-support measures is not a matter of a few isolated corner cases, but of progressively reweighting a wide interior distribution.

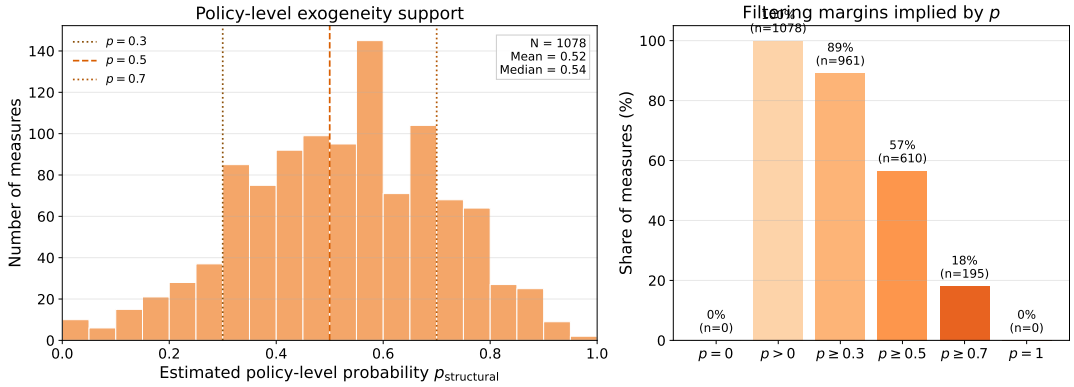

Figure 7: Distribution of Policy-Level Exogeneity Support for U.S. Budget Tax Measures

Note: The figure uses the budget-based U.S. tax measures with complete implementation dates, amounts, and policy-level structural probabilities that can be matched to the quarterly macro sample. The left panel plots the distribution of p, with vertical reference lines at 0.3, 0.5, and 0.7. The right panel reports the share of measures satisfying selected thresholds. In the current sample, there are no exact corner observations at $p = 0$ or $p = 1$.

5.2.4 From estimated support to quarterly tax shocks

Figure 8 translates the policy-level support measure into quarterly tax shocks. The left panel compares three series: all budget tax changes treated symmetrically, the corresponding soft-p weighted budget shock, and the original Romer–Romer exogenous tax shock series. The weighting step is economically consequential. It leaves the budget series active in many quarters, but it sharply compresses its total absolute shock mass, indicating that the resulting series is not simply a rescaled version of the raw budget instrument.

The right panel makes this point directly. Relative to the unfiltered budget series, the soft-p construction retains far less total shock mass while remaining much denser than the RR benchmark. This is the intended role of the end-to-end procedure: preserve the additional temporal support contributed by routine budget measures, but down-weight the component of that support that receives weaker documentary backing for exogeneity.

5.2.5 Macroeconomic comparison with the Romer–Romer benchmark

For the end-to-end document exercise, I return to the paper's baseline estimator rather than forcing the newly constructed shocks through the exact Romer–Romer distributed-lag replication. Specifically, I estimate horizon-by-horizon local projections of log GDP on the contemporaneous tax shock, four lags of the shock, and four lags of log GDP, with 68% Newey–West confidence bands. The original RR exogenous shock series is passed through that same local-projection specification, so the comparison below is an apples-to-apples comparison of shock constructions under the common methodology used in the main U.S. application.

Figure 9 reports the three budget-based constructions that are most informative in this baseline LP

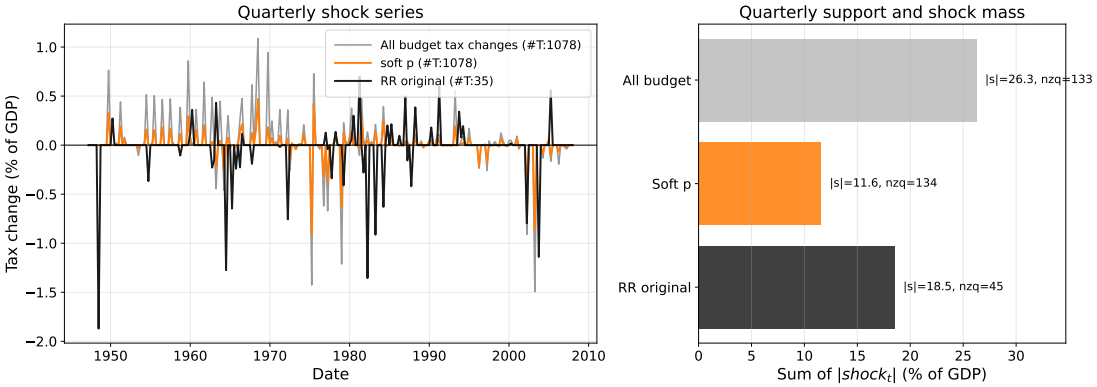

Figure 8: From Budget Measures to Quarterly Tax Shocks

Note: The left panel plots quarterly tax-shock series, all expressed as a percentage of contemporaneous GDP: all budget tax changes, the soft-p weighted budget shock, and the original Romer–Romer exogenous tax shock series. The right panel reports diagnostic aggregates for those same constructions. "nzq" denotes the number of non-zero quarters and $|s|$ denotes the sum of absolute quarterly shocks over the matched macro sample.

setup. The left panel applies a hard p-filter at different $\bar{p}$. Under the local-projection estimator, this high-support series is more contractionary than the RR benchmark over medium horizons. The middle panel uses soft weighting. That specification is also contractionary and, around the peak response, is closer in magnitude to the RR benchmark than the hard-threshold series. The right panel reports the 2-type $p = 1$ endpoint. This endpoint is likewise contractionary and remains close to the RR benchmark through the medium run, while becoming more negative at longer horizons.

The relevant end-to-end conclusion is therefore not that the new budget-based shocks replicate Romer and Romer one-for-one. It is that documentary support extracted from policy documents materially reshapes the budget-based series, and that once the shocks are estimated with the same baseline LP methodology used elsewhere in the paper, all three high-support constructions generate sizable contractionary responses. Appendix E.4 reports two related robustness exercises: a direct comparison between filtered and unfiltered tax shocks in both frameworks, and an augmentation exercise in which newly-identified budget shocks are added outside the original RR sample. In that latter exercise, point estimates move little, suggesting that the additional budget measures primarily expand support rather than overturn the benchmark.

Taken together, the U.S. document exercise should be read as an out-of-sample extension of the RR identification logic. The exercise does not start from hand-written RR narratives and it does not attempt to reproduce the original episode list. Instead, it asks whether the SANI pipeline can recover economically meaningful identifying variation when applied to a denser universe of routine policy measures and their contemporaneous documentation. The answer from the benchmark environment is positive, but nuanced: documentary support clearly improves on the raw-budget series, and continuous weighting appears more informative than a hard binary filter, yet the resulting budget-based shocks remain weaker than the canonical RR series.

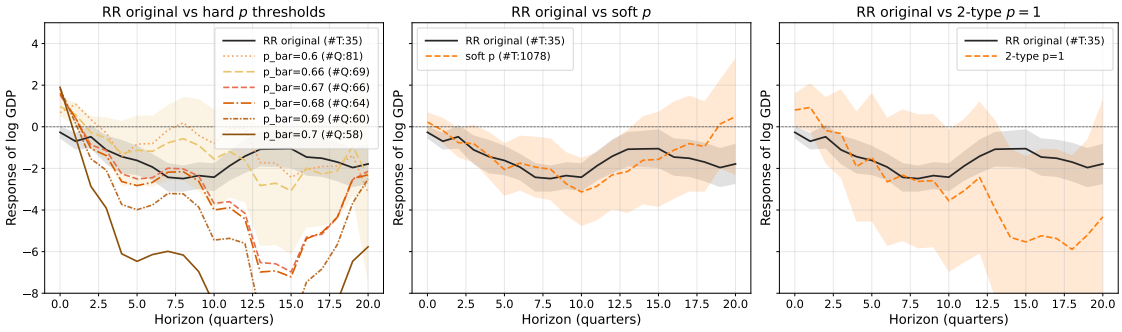

Figure 9: Romer–Romer Benchmark vs. Newly Constructed Budget-Based Tax Shocks

Note: Each panel reports responses of log GDP estimated with the baseline local-projection specification used in the action-level U.S. application: contemporaneous shock, four lags of the shock, four lags of log GDP, and 68% Newey–West confidence bands. The black curve is the original Romer–Romer exogenous tax shock series estimated under that same LP specification. The orange dashed curve is a budget-based construction obtained from the end-to-end SANI procedure: the hard-threshold shock with $\bar{p} = 0.67$ in the left panel, the soft-p weighted shock in the middle panel, and the 2-type $p = 1$ endpoint in the right panel. For visual comparison only, the left panel also overlays point estimates for hard-threshold shocks from $\bar{p} = 0.6$ to $\bar{p} = 0.7$, without additional confidence bands.

6 Conclusion

This paper revisits narrative identification in macroeconomics and argues that its core logic naturally admits a graded interpretation. Traditional narrative designs rely on binary inclusion rules that treat selected policy actions as fully exogenous while discarding the remainder. This paper generalizes that approach by allowing qualitative documentary evidence to enter continuously, reallocating identifying weight across policy actions according to the strength of narrative support for orthogonality in expectation. Identification is recast at the level of reduced-form moments, exploiting systematic variation in narrative support without requiring the recovery of observation-level structural policy shocks.

To operationalize this perspective, the paper develops a document-grounded implementation that translates policymakers' stated motivations into a continuous index of narrative support for exogeneity. The implementation systematizes the use of historical sources while preserving traceability to the underlying documentary record and incorporating diagnostics that discipline how textual evidence contributes to identification. These features are designed to support, rather than replace, economic judgment, and to make explicit the evidentiary basis underlying narrative restrictions.

The empirical analysis serves two complementary purposes. First, the paper provides an external validation anchored in the canonical narrative tax series of Romer and Romer (2010). Using a newly constructed dataset of routine U.S. federal tax measures drawn from the annual budget process, high-support budget-based shocks estimated under the paper's baseline local-projection setup generate sizable contractionary responses that are economically comparable to the Romer–Romer benchmark, even though the ranking of thresholded, soft-weighted, and two-type constructions depends on the estimator used. Augmenting the original narrative series with these additional measures primarily improves precision by expanding coverage, without materially altering the implied macroeconomic effects. Second, the paper constructs a new long-run narrative dataset of French value-added tax policy from 1960 to 2024 and documents the dynamic macroeconomic effects of VAT changes. VAT increases with strong narrative support for exogeneity are persistently contractionary, with a short-

lived increase in inflation consistent with tax pass-through, highlighting the distinct transmission of indirect taxes.

More broadly, the framework clarifies how narrative evidence can discipline causal inference without requiring sharp exogeneity claims at the level of individual policy actions. By relaxing the implicit zero-contamination assumption underlying binary classifications, graded narrative identification retains information from a wider range of policy interventions while preserving the interpretive foundations of narrative approaches. This perspective helps address the sparsity of traditional narrative series and provides a principled way to trade off credibility and statistical power.

The analysis also illustrates how large collections of policy documents can be incorporated systematically into empirical macroeconomics when combined with transparent, document-grounded procedures. By lowering the cost of organizing and interpreting qualitative evidence, the framework expands the feasibility of narrative designs in settings characterized by frequent, lower-salience policy actions. Future work can build on this foundation by developing estimators that directly exploit variation in narrative support under weaker orthogonality assumptions and by extending the approach to other domains of fiscal, monetary, and structural policy.

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7 Appendix

A Mathematical Model for Narrative Identification

This appendix collects the formal results underpinning Section 1. It (i) states the causal boundary object and the identifying restriction, (ii) derives the population estimands of weighted local projections based on graded narrative support, and (iii) shows how two-types local projections deliver a parametric boundary estimator together with a simple nesting diagnostic. Throughout, all moments are understood to be conditional on predetermined controls Z_t as in the main text; equivalently, interpret all variables below after Frisch–Waugh–Lovell residualization with respect to Z_t.

A.1 A. Boundary object and identification

Recall the support-indexed moments defined in (4):

$$\mu_h(p) \equiv \mathbb{E}(y_{t+h}\,\Delta\tau_{j,t} \mid p_{j,t} = p)\,, \qquad v(p) \equiv \mathbb{E}\big(\Delta\tau_{j,t}^2 \mid p_{j,t} = p\big)\,,$$

and the support-indexed response function (**??**):

$$r_h(p) \equiv \frac{\mu_h(p)}{v(p)} \qquad (v(p) > 0).$$

The graded narrative target is the boundary response

$$r_h(1) \equiv \lim_{p\to 1} r_h(p),$$

interpreting the limit along the upper tail of narrative support when the empirical support does not place mass at $p = 1$.

Assumption .1 (Boundary orthogonality (alignment))**.** For each horizon $h \geq 0$,

$$\lim_{p\to 1} \mathbb{E}(\varepsilon_{t+h}\,\Delta\tau_{j,t} \mid p_{j,t} = p) = 0, \tag{24}$$

where ε_{t+h} is the disturbance in the structural relation (1).

Proposition .2 (Boundary causal interpretation)**.** *Under Assumption .1 and continuity of the relevant conditional moments,*

$$r_h(1) = \beta_h.$$

Proof. From (1) and the definition of $\mu_h(p)$,

$$\mu_h(p) = \mathbb{E}((\beta_h \Delta\tau_{j,t} + \varepsilon_{t+h})\,\Delta\tau_{j,t} \mid p_{j,t} = p) = \beta_h\, v(p) + \mathbb{E}(\varepsilon_{t+h}\Delta\tau_{j,t} \mid p_{j,t} = p)\,.$$

Divide by $v(p)$ to obtain

$$r_h(p) = \beta_h + \frac{\mathbb{E}(\varepsilon_{t+h}\Delta\tau_{j,t} \mid p_{j,t} = p)}{v(p)}.$$

Taking $p \to 1$ and applying (24) yields $r_h(1) = \beta_h$. □

A.2 B. weighted local projections as weighted functionals of $r_h(p)$

Given a weighting rule $g : [0,1] \to \mathbb{R}$, the paper constructs the time-t policy measure

$$s_t(g) \equiv \sum_j g(p_{j,t})\, \Delta\tau_{j,t}. \tag{25}$$

The weighted local projection is (7):

$$y_{t+h} = \alpha_h + \beta_h(g)\, s_t(g) + \Gamma_h' X_t + u_{t+h}(g).$$

The results below are stated in the time-unit notation to streamline the derivations: let $\Delta\tau_t$ and p_t denote the corresponding period-t aggregates (or a single-action environment), and define $s_t(g) = g(p_t)\Delta\tau_t$. In the empirical implementation in the main text, $s_t(g)$ is defined by (25); all statements below should be interpreted as characterizing the estimand of a one-regressor LP based on a generic policy measure $s_t(g)$, with the role of $g(\cdot)$ being to shift weight toward high-support variation.

Define the conditional moments (suppressing Z_t):

$$v(p) \equiv \mathbb{E}\big((\Delta\tau_t)^2 \mid p_t = p\big), \qquad \mu_h(p) \equiv \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p), \qquad r_h(p) \equiv \frac{\mu_h(p)}{v(p)}.$$

Lemma .3 (Proxy LP estimand)**.** *Consider the population regression of y_{t+h} on $s_t(g) = g(p_t)\Delta\tau_t$ (after partialling out predetermined controls). The population coefficient satisfies*

$$\beta_h(g) = \frac{\mathbb{E}(y_{t+h} s_t(g))}{\mathbb{E}(s_t(g)^2)} = \frac{\mathbb{E}(g(p_t)\, \mu_h(p_t))}{\mathbb{E}(g(p_t)^2\, v(p_t))} = \frac{\mathbb{E}(g(p_t)\, v(p_t)\, r_h(p_t))}{\mathbb{E}(g(p_t)^2\, v(p_t))}. \tag{26}$$

Proof. By definition of the population regression coefficient,

$$\beta_h(g) = \frac{\mathbb{E}(y_{t+h} s_t(g))}{\mathbb{E}(s_t(g)^2)}.$$

Substitute $s_t(g) = g(p_t)\Delta\tau_t$ and apply iterated expectations conditional on p_t:

$$\mathbb{E}(y_{t+h} s_t(g)) = \mathbb{E}(g(p_t)\, \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t)) = \mathbb{E}(g(p_t)\mu_h(p_t)),$$

and

$$\mathbb{E}(s_t(g)^2) = \mathbb{E}\big(g(p_t)^2\, \mathbb{E}((\Delta\tau_t)^2 \mid p_t)\big) = \mathbb{E}(g(p_t)^2 v(p_t)).$$

Using $\mu_h(p) = v(p) r_h(p)$ yields (26). □

Lemma .3 makes clear that weighted local projections generally estimate a weighted functional of $r_h(p)$ over the empirical support of p_t, rather than the boundary object $r_h(1)$.

A.3 C. A unified bias representation

Write

$$r_h(p) = r_h(1) + \Delta_h(p), \qquad \Delta_h(p) \equiv r_h(p) - r_h(1).$$

For weighted local projections based on $s_t(g) = g(p_t)\Delta\tau_t$, the proxy estimand can be decomposed into a deviation term (reflecting the interior behavior of $r_h(p)$) and a purely mechanical scale wedge induced by the choice of $g(\cdot)$.

Proposition .4 (Decomposition relative to the boundary target)**.** *Under the conditions of Lemma .3,*

$$\beta_h(g) - r_h(1) = \frac{\mathbb{E}(g(p_t)\, v(p_t)\, \Delta_h(p_t))}{\mathbb{E}(g(p_t)^2\, v(p_t))} + r_h(1)\left(\frac{\mathbb{E}(g(p_t)\, v(p_t))}{\mathbb{E}(g(p_t)^2\, v(p_t))} - 1\right). \tag{27}$$

Proof. By Lemma .3,

$$\beta_h(g) = \frac{\mathbb{E}(g(p_t)\, v(p_t)\, r_h(p_t))}{\mathbb{E}(g(p_t)^2\, v(p_t))}.$$

Substitute $r_h(p_t) = r_h(1) + \Delta_h(p_t)$ and rearrange to obtain (27). □

Normalization for comparability (optional). Because $s_t(g) = g(p_t)\Delta\tau_t$ changes with $g(\cdot)$, the coefficient $\beta_h(g)$ is in "units" of the proxy and is not directly comparable across weighting rules. A convenient normalization is

$$\tilde{\beta}_h(g) \;\equiv\; \beta_h(g) \cdot \frac{\mathbb{E}\big(g(p_t)^2\, v(p_t)\big)}{\mathbb{E}(g(p_t)\, v(p_t))} = \frac{\mathbb{E}(g(p_t)\, v(p_t)\, r_h(p_t))}{\mathbb{E}(g(p_t)\, v(p_t))}. \tag{28}$$

Then

$$\tilde{\beta}_h(g) - r_h(1) = \frac{\mathbb{E}(g(p_t)\, v(p_t)\, \Delta_h(p_t))}{\mathbb{E}(g(p_t)\, v(p_t))}, \tag{29}$$

which isolates the role of $g(\cdot)$ in shifting weight toward high-support observations.

Remark (bias–variance tradeoff). Expression (27) (or, after normalization, (29)) formalizes the bias–variance tradeoff emphasized in the main text. Choosing $g(\cdot)$ to concentrate weight near $p = 1$ reduces the deviation term under alignment (to the extent that $\Delta_h(p) \to 0$ as $p \to 1$), but it also typically reduces the effective signal $\mathbb{E}(g(p_t)^2 v(p_t))$ and increases sampling variability. The scale wedge in (27) vanishes for hard thresholds, where $g^2 = g$.

A.4 D. Two-types local projections as parametric boundary estimators

The two-types local projection in (8) is

$$y_{t+h} = \alpha_h + \theta_h\, \Delta\tau_t + \lambda_h\, (p_t \Delta\tau_t) + \Gamma_h' Z_t + \upsilon_{t+h}. \tag{30}$$

Equivalently,

$$y_{t+h} = (\theta_h + \lambda_h p_t)\Delta\tau_t + \Gamma_h' Z_t + \upsilon_{t+h}.$$

This specification corresponds to the affine approximation

$$r_h(p) \approx \theta_h + \lambda_h p, \tag{31}$$

under which the boundary object equals

$$r_h(1) = \theta_h + \lambda_h.$$

Accordingly, the implied boundary estimator in the paper is

$$\widehat{r}_h(1) \equiv \hat{\theta}_h + \hat{\lambda}_h.$$

A.5 E. Nesting and a diagnostic test

Define

$$w_t \equiv p_t \Delta\tau_t, \qquad \tilde{b}_t \equiv (1-p_t)\Delta\tau_t, \qquad \Delta\tau_t = w_t + \tilde{b}_t.$$

Rewrite (30) as

$$y_{t+h} = \alpha_h + (\theta_h + \lambda_h)\, w_t + \theta_h\, \tilde{b}_t + \Gamma_h' Z_t + \upsilon_{t+h}. \tag{32}$$

Proposition .5 (Nesting of the soft proxy LP)**.** *The weighted local projection using the soft proxy $w_t = p_t \Delta\tau_t$ is nested in the two-types LP* (32) *as the restriction $H_0 : \theta_h = 0$.*

Proof. Imposing $\theta_h = 0$ in (32) yields

$$y_{t+h} = \alpha_h + \lambda_h w_t + \Gamma_h' Z_t + \upsilon_{t+h},$$

which is the soft proxy LP (up to notation). □

Testing $H_0 : \theta_h = 0$ provides a diagnostic for whether compressing graded narrative information into the single proxy w_t is empirically adequate at horizon h.

B A Document-Grounded Implementation of Graded Narrative Identification

B.1 Illustration of the mapping

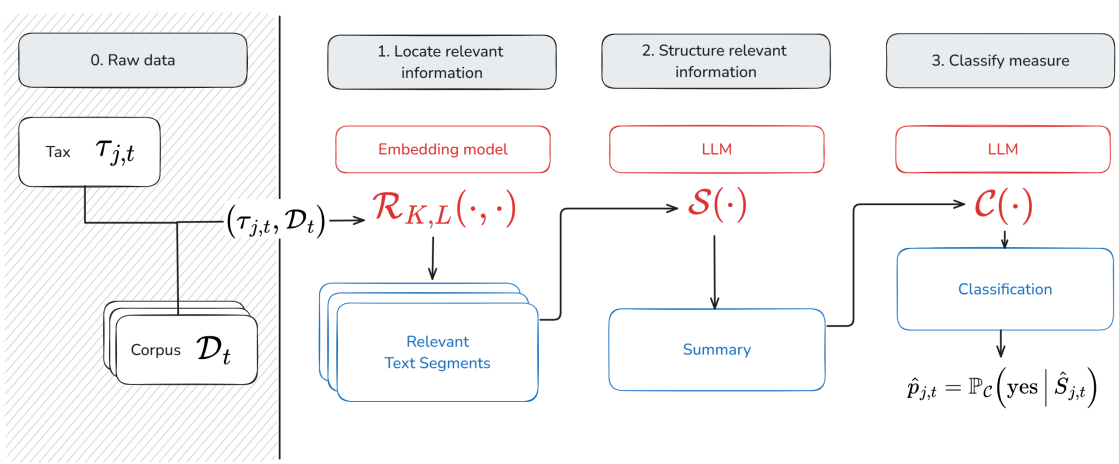

Figure 10: Mapping from policy descriptions and contemporaneous documents to narrative classifications. The figure illustrates the sequence $(p_{j,t}, \mathcal{D}_t) \to \hat{R}_{j,t} \to \hat{S}_{j,t} \to \hat{c}_{j,t}$ defined in Section 2.

B.2 Embedding Model and Distance Metric

Semantic retrieval relies on vector representations of text segments and policy descriptions generated by a transformer-based embedding model. In the baseline implementation, embeddings are obtained using OpenAI's `text-embedding-3-large` model, which maps each text input to a 3,072-dimensional dense vector in $\mathbb{R}^{3072}$. The model is trained to capture semantic relationships between texts such that semantically similar passages are represented by vectors with small angular separation.

For each policy description p_j and document segment $d_k \in \mathcal{D}$, the corresponding embedding vectors q_j and v_k are normalized to unit length. Semantic proximity is then evaluated using the cosine distance,

$$s_{j,k} = 1 - \frac{q_j \cdot v_k}{\|q_j\| \, \|v_k\|},$$

so that smaller values of $s_{j,k}$ indicate greater semantic similarity. Semantic proximity between text embeddings is measured using the cosine distance. Because all embeddings are normalized to unit length, cosine and Euclidean distances are mathematically equivalent in this setting, yielding identical retrieval rankings under either metric. Distances are computed through standard nearest-neighbor search routines provided by the underlying vector store, which efficiently identifies the text segments closest to each policy query in the embedding space. This normalization ensures that retrieval reflects semantic similarity rather than differences in vector magnitude. The retrieval operator defined in Equation (10) in the main text therefore selects, for each policy event j, the K document segments with the smallest cosine distances to the corresponding policy query q_j.

B.3 Example details

Applying the retrieval operator $\mathcal{R}_{K,L}$ to the query $q_{j,t}$ and corpus $\mathcal{D}_t$, I obtain the top-K most relevant passages. Here are the top-5 passages:

- **Chunk 1.** On December 31, 1976, to mitigate, by a reduction of the weight of taxes bearing on consumption, the upward movement likely to occur at the end of this phase of price freeze; — it constitutes the last stage of a reform undertaken since 1969 — the date on which the standard rate of VAT was 23.456% — which results, ultimately, in a reduction, in less than seven years, of one quarter of the standard rate and in the merging of the latter with the intermediate rate.

- **Chunk 2.** An exceptional contribution would be requested in 1977 from the wealthiest French; it would affect all of them and only them, contrary to what had been provided for in the deleted article 5. This taxation would have as its base the total of the bases corresponding to the elements of lifestyle enumerated in article 168 of the General Tax Code, with the exception of main residences and cars of a power equal to or less than 16 horsepower. The income-equivalent threshold, calculated according to the norms of the aforementioned article 168, above which the exceptional contribution applies, is set at 60,000 F and the number of accumulated elements provoking taxation is three. Your Finance Committee proposes that you adopt the present article. VAT (Value Added Tax) – Article 6. Reduction of the standard rate and unification of the standard rate and the intermediate rate. Text. — I. — The standard rate of value-added tax is set at 17.60%. In the departments of Guadeloupe, Martinique and Réunion, it is set at 7.50II. — The upper limit of the special rebate for artisans fixed in paragraph 3 of article 282 of the General Tax Code is raised to 20,000 F. III. — The provisions of the present article enter

into force on January 1, 1977. Comments. — It is proposed in the present article to lower by 2.4 points, starting from January 1, 1977, the standard rate of value-added tax which is currently 20% and, thereby, to bring it down to the level of the intermediate rate, namely 17.60%. This measure fits within a double perspective: — it is part of the program of combating inflation and is of a nature to permit, after the price freeze applied during the period from September 15 to December 31, 1976, to mitigate, by a reduction of the weight of taxes bearing on consumption, the upward movement likely to occur at the end of this phase of price freeze;

- **Chunk 3.** The number of rates applied in the matter of value-added tax, starting from next January 1, will be reduced to three instead of four; there would remain only: — the increased rate (33 %: cars, motorcycles, photography and cinematography equipment, sound and image recording devices, precious stones and precious metals, etc.); — the standard rate (17.6%: energy products, beverages, services and all transactions hitherto subject to the 20% rate, that is to say all those which were not taxed at the increased rate, nor at the intermediate rate, nor at the reduced rate); — the reduced rate (7%: essentially food products, pharmaceutical products, water and fertilizers). At the same time, it is planned to carry out a reduction of 2.5 points of the standard VAT rate and to align it with the intermediate rate in the departments of Guadeloupe, Martinique and Réunion, French Guiana not being concerned since the value-added tax regime is not applied there. In the three departments concerned, where, due to constraints inherent in the economies of developing countries, VAT rates are significantly lower than those of Metropolitan France, starting from January 1, 1977, one will record: — the increased rate: 14%; — the standard rate: 7.5%; — the reduced rate: 3.5%. In order to support the artisanal professions, it is proposed in paragraph II of the present article to raise, starting from January 1, 1977, the upper limit of the special rebate from which they benefit, raising it from 13,500 F to 20,000 F. Let us recall that, in order to lighten the burdens of small taxpayers, and in particular artisans, article 282 of the General Tax Code has provided, in their favor, special provisions resulting in a waiver or a reduction of the VAT normally incumbent upon them.

- **Chunk 4.** Standard of 6.5% per year. b) The elimination of the levy. As had been provided by the Finance Law for 1976, in the initial text presented by the Government, the levy was to cease to apply when the progression of prices was below a set norm: this one, in the governmental project, was fixed at 2.50% for the consumer price index of the group of manufactured products during a period of six consecutive months.

- **Chunk 5.** Article: — in paragraph I, the objective is to encourage business leaders to convert into capital the advances they are led to grant in the form of deposits in current accounts to the companies they manage. Up to now, in such cases, the ordinary contribution right was a proportional duty of 1%; it is planned to substitute for it a fixed duty of 220 F on the condition, however, that the sums thus incorporated into the capital have been placed at the constant disposal of the company, for a minimum period of twelve months.

B.4 Technical Implementation Details

The deterministic implementation relies on several technical specifications to ensure reproducibility. Both $\mathcal{S}$ and $\mathcal{C}$ operators use fixed prompt templates with the following decoding parameters: tempera-

ture = 0, top-p = 1.0, no nucleus sampling, and fixed maximum token limits. For transformer models, this eliminates stochastic sampling during generation.

For open-source models with frozen weights (e.g., LLaMA, Mistral family), deterministic decoding combined with identical computational environments guarantees bitwise-identical outputs across runs. This provides the strongest form of reproducibility, as the complete model architecture, weights, and inference code are publicly auditable.

For proprietary APIs (e.g., OpenAI GPT models, Google Gemini), additional logging is implemented to track potential sources of non-reproducibility. Each API call logs: (i) the specific model identifier and version, (ii) SHA-256 hashes of the prompt template and input content, (iii) all decoding parameters, (iv) timestamp and API response metadata, and (v) the complete generated output. This comprehensive logging enables detection of silent model updates or API changes that could affect results.

Even with deterministic settings, proprietary models may exhibit minor non-determinism due to distributed inference infrastructure, floating-point precision differences, or undocumented system changes. Additionally, model providers may silently update their systems, potentially altering outputs for identical inputs. While such changes are typically small, they can accumulate across large-scale classification tasks. The metadata logging system provides an audit trail that enables researchers to verify whether results were generated under identical conditions and to detect potential sources of variation in replication attempts.

B.5 Interpretation of the Rescaling Transformation

The normalization in equation (**??**) admits a simple affine interpretation. Suppose that for each category c the raw logits satisfy

$$Y_{\ell,c} = a_c + b_c \theta_{\ell,c}, \qquad b_c > 0,$$

where $\theta_{\ell,c}$ is a latent semantic intensity measuring the alignment of motivation M_ℓ with category c, and (a_c, b_c) capture prompt-specific location and scale distortions. Let $\theta_{\min}$ and $\theta_{\max}$ denote the latent intensities corresponding to the negative and positive reference prompts. Then

$$\tilde{Y}_{\ell,c} = \frac{Y_{\ell,c} - Y_{\min,c}}{Y_{\max,c} - Y_{\min,c}} = \frac{\theta_{\ell,c} - \theta_{\min}}{\theta_{\max} - \theta_{\min}}.$$

The rescaled score therefore removes prompt-specific distortions (a_c, b_c) and recovers the latent semantic intensity up to a common normalization. The construction does not require the affine structure to hold exactly; it is sufficient that the mapping from latent semantic alignment to logits be strictly monotone within each category.

C Grounding Diagnostics: Marginal Narrative Contributions

C.1 Robustness of the Linear Surrogate Estimator

Because the surrogate model is estimated with $n = 64$ ablations and up to $K \approx 30$ retrieved passages, I assess whether the resulting coefficients are sensitive to the choice of regularisation. Ridge regression yields estimates that are nearly identical to the OLS baseline and improves numerical stability when

inclusion patterns exhibit collinearity. A LASSO with $\lambda = 0.01$ selects the same high-weight passages as OLS and produces coefficients of similar magnitude. These results indicate that the dependence patterns reported in the main text are not driven by estimator choice or overfitting. For settings with substantially larger K, penalised estimators would provide a natural extension.

C.2 Global Placebo Test for Summarisation Grounding

The coefficient-level placebo test in the main text evaluates, for each passage individually, whether its positive marginal contribution is unusually large relative to the distribution of positive placebo contributions. Because this diagnostic is applied to multiple passages within each event, some significant cells may be mechanically expected under the null of no grounding. This appendix reports an omnibus test that aggregates evidence across passages and verifies that the overall pattern of significant contributions cannot be attributed to chance.

Placebo mass. For each event, let $w_k^+ = \max\{\hat{w}_k, 0\}$ denote the positive marginal contribution of passage k, and let R and P denote the sets of relevant and placebo passages, respectively. The share of positive weight falling on placebo passages is

$$\phi = \frac{\sum_{k \in P} w_k^+}{\sum_{k \in R \cup P} w_k^+} \in [0, 1],$$

which summarises, at the event level, how much of the model's total positive dependence is placed on semantically irrelevant text. Grounded behaviour is characterised by low ϕ, while ungrounded dependence yields large values.

Randomisation inference. To assess whether the observed value ϕ_{obs} is unusually large, I test the sharp null that placebo labels are irrelevant to the pattern of marginal contributions. Let K^+ denote the number of passages with positive weights. Under the null, any assignment of K^+ placebo labels across passages is equally likely. I therefore compute the randomisation distribution by permuting placebo labels across the K^+ positions, recomputing placebo mass ϕ_b for each permutation, and forming the right-tailed p-value:

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{\phi_b \geq \phi_{\text{obs}}\}}{1 + B}.$$

A small p-value indicates that the model places significantly more positive weight on placebo passages than expected under random assignment, implying ungrounded behaviour. Conversely, when ϕ_{obs} falls well within the reference distribution, the observed coefficient-level significance pattern is unlikely to arise from multiple testing alone.

Interpretation. This global test complements the passage-level diagnostic in the main text. While the main procedure identifies *which* passages provide statistical support for each motivation, the global placebo mass test assesses *whether the overall pattern of support is itself meaningful.* Across events, the observed placebo masses cluster well below their randomisation-based critical values, confirming that the significant passages identified in the main text do not arise from chance alone.

D Application of SANI

D.1 Retrieval Performance Metrics and Parameter Selection [TO BE ADDED]

This Appendix provides additional detail on the evaluation of the retrieval operator $\mathcal{R}_{K,L}$, the calibration of its parameters (K, L), and the performance of alternative segmentation strategies. The objective is to document the robustness of retrieval accuracy and to clarify how the parameter settings used in the main text were chosen.

Benchmark Setup Each VAT measure (j, t) is manually linked to its corresponding Senate report article, denoted the *gold article*. For every candidate pair of parameters (K, L), I perform retrieval and record all text segments d_k whose span overlaps the gold article. These are counted as *hits*. Because each article can extend across multiple segments, a single event may have several true positives.

D.2 Classification Prompts and Category Scheme

This appendix reports the prompt templates used to classify value-added tax (VAT) measures into categories following the narrative taxonomy of Cloyne (2013). Each category corresponds to one of the subtypes summarized in Table 1. The classification operator $\mathcal{C}$ applies these prompts independently to the structured motivations $\hat{S}_{j,t}$ obtained from the summarization stage, returning binary (*Yes/No*) assessments and a textual rationale grounded in the retrieved evidence.

Prompt templates

Each prompt is presented in a standardized format and corresponds to one of the classification categories listed in Table 1. The prompts are organized by group—*endogenous* (Demand Management 3, Supply Stimulus 4, Deficit Reduction 5, Spending-driven 6) and *exogenous* (Long-term 7, Ideological 8, External 9, Deficit Consolidation 10). Each prompt provides the task definition, inclusion and exclusion criteria, and output format used in the classification procedure. Together, they operationalize the decision rules summarized in Table 1, ensuring full transparency and reproducibility of the classification process.

English translation of the classification prompt (Demand Management / DM):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Demand Management (DM)"?

Definition of Demand Management (DM):
A passage is classified as Demand Management if the measure is motivated by the need to influence, in the short term, the level or composition of aggregate demand in the economy—either to curb inflationary pressures and cool overheated sectors, or to stimulate demand during a slowdown or recession.

This includes:

- Tax increases or restrictions explicitly justified as reducing consumption, restraining demand, or mitigating inflationary pressures.
- Tax cuts or reliefs explicitly presented as stimulating demand, consumption, investment, or economic activity.
- Measures related to concerns over the balance of payments, inflation control, or unemployment, where the mechanism operates through aggregate demand.
- Temporary or reversible measures (e.g., surcharges, freezes, deferrals) justified by immediate economic circumstances, particularly short-term demand conditions.

*Do **not** classify as Demand Management if:*

- The motivation is solely deficit reduction or fiscal consolidation.
- The measure is driven exclusively by environmental, health, or equity objectives without reference to aggregate demand.
- The change is a routine indexation or neutral adjustment without intent to influence demand.
- The measure is presented as a long-term structural reform (improving incentives or competitiveness) rather than targeting short-term demand effects.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 3: Classification Prompt for the "Demand Management (DM)" Category

English translation of the classification prompt (Supply stimulus / SS):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Supply stimulus (SS)"?

Definition of Supply stimulus (SS):
A passage is classified as Supply stimulus if the measure is motivated by the need to stimulate, in the short term, production, investment, competitiveness, or employment on the supply side of the economy. Such measures typically involve tax cuts, reliefs, or structural changes aimed at improving incentives, profitability, liquidity, or business conditions to increase output and growth potential.

This includes:

- Reductions in income tax rates, allowances, or thresholds explicitly presented as improving work incentives or effort.
- Corporate or business tax cuts justified as encouraging investment, profitability, or competitiveness.
- Investment allowances or reductions in social contributions intended to lower hiring costs or promote employment.
- Measures described as supporting recovery or short-term growth through improvements on the supply side.
- Cyclical adjustments designed to maintain the immediate competitiveness of a sector.

*Do **not** classify as Supply stimulus if:*

- The justification emphasizes long-term structural, productivity, or competitiveness effects.
- The measure is presented as deficit reduction or fiscal consolidation.
- The measure targets aggregate demand management (short-term demand stimulation or restraint).
- The justification is primarily environmental, health-related, or social without a supply-side argument.
- The change is a neutral indexation, a temporary demand stimulus, or a routine adjustment without an incentive-based rationale.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 4: Classification Prompt for the "Supply stimulus (SS)" Category

English translation of the classification prompt (Deficit Reduction / DR):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Deficit Reduction (DR)"?

Definition of Deficit Reduction (DR):
A measure is classified as Deficit Reduction if its objective is to generate immediate revenue or prevent a loss of revenue in order to contain the short-term budget deficit.

This includes:

- The freezing or restriction of tax allowances or reliefs to raise funds for the national treasury "this year" or "next year."
- Increases in taxes or duties explicitly presented as necessary to raise additional revenue (e.g., "I need to raise X in additional receipts," "to finance this," "given the need to generate extra revenue").
- The elimination or reduction of costly reliefs whose expense is deemed unsustainable or unjustified.
- Adjustments to consumption taxes—such as fuel, alcohol, tobacco, or VAT—to "protect real yields" or prevent erosion of the tax base.
- Temporary or exceptional levies introduced to cover revenue needs within the current fiscal year.

*Do **not** classify as Deficit Reduction if:*

- The measure is primarily motivated by medium- or long-term fiscal consolidation (this belongs to another category).
- The measure is justified mainly by non-fiscal motives (environmental, health, or social equity) without clear reference to deficit control.
- The measure reflects routine indexation or neutral adjustments that do not raise revenue.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 5: Classification Prompt for the "Deficit Reduction (DR)" Category

English translation of the classification prompt (Spending-driven / SD):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Spending-driven (SD)"?

Definition of Spending-driven (SD):
A passage is classified as Spending-driven if the measure is explicitly motivated by the need to finance, offset, or enable increases in public spending or the introduction of new expenditure programs.

This includes:

- Tax increases or restrictions on reliefs introduced to cover the cost of new spending or the expansion of existing programs (e.g., food subsidies, NHS funding, social reforms).
- Statements indicating that additional revenue is needed to "finance," "help finance," or "enable" proposed spending measures.
- Exceptional or one-off taxes whose purpose is to fund a specific expenditure program (e.g., employment initiatives, child poverty measures).
- Reallocation or withdrawal of tax reliefs to free resources for targeted spending programs.

*Do **not** classify as Spending-driven if:*

- The motivation is deficit reduction or fiscal consolidation for long-term sustainability.
- The measure is justified by demand management (inflation control, balance-of-payments adjustment, or demand stimulus).
- The change is primarily motivated by environmental, health, or equity considerations without explicit reference to spending needs.
- The measure is a routine indexation or neutral adjustment with no link to expenditure financing.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 6: Classification Prompt for the "Spending-driven (SD)" Category

English translation of the classification prompt (Long-term / LR):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Long-term (LR)"?

Definition of Long-term (LR):
A passage is classified as Long-term if the measure is motivated by structural, supply-side, or long-term incentive effects on the economy—aimed at improving productivity, competitiveness, investment, saving, or economic efficiency over time.

This includes:

- Tax cuts or reforms justified by improving work effort, incentives, or morale.
- Measures designed to stimulate business investment, capital formation, or technological modernization.
- Reforms targeting international competitiveness, efficiency, or simplification of the tax system.
- Policies encouraging entrepreneurship, risk-taking, or saving to support future growth.
- Changes presented as necessary for the transition to a new economic regime (e.g., from wartime to peacetime).

*Do **not** classify as Long-term if:*

- The main motivation is short-term revenue needs or deficit consolidation.
- The measure concerns only short-term demand management (stimulus or demand restraint).
- The justification is purely environmental, health-related, or equity-driven without reference to long-term incentives or growth.
- The change is a routine indexation or minor administrative adjustment with no structural intent.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 7: Classification Prompt for the "Long-term (LR)" Category

English translation of the classification prompt (Ideological / IL):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Ideological (IL)"?

Definition of Ideological (IL):
A passage is classified as Ideological if the measure is primarily motivated by political values, social philosophy, principles of fairness, or a broader vision of society and the tax system—rather than by short-term macroeconomic management (demand control) or fiscal balance (deficit reduction or consolidation).

This includes:

- References to fairness, justice, tradition, or "social values," including environmental or ethical considerations.
- Tax reliefs or exemptions targeted at "deserving" groups (e.g., retirees, widows, children, families, vulnerable populations).
- Explicitly moral, social, or political justifications (e.g., "people hate income tax more than any other tax," "to reward work," "to encourage saving," "to support family unity").
- Long-term reforms aimed at making the tax system simpler, fairer, or more logical, without references to deficit or demand management.
- Measures presented as fulfilling electoral promises or party commitments (e.g., VAT cuts on essential goods, new redistributive taxes).
- Reforms introduced as part of a broader vision of taxation reflecting social or political values (e.g., taxing capital gains "as income," inheritance thresholds for family fairness).

*Do **not** classify as Ideological if:*

- The measure is primarily motivated by deficit reduction, fiscal sustainability, or budget consolidation.
- The change is framed in terms of short-term demand management (stimulus, inflation control, balance of payments).
- The measure is a routine indexation or a neutral technical adjustment.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 8: Classification Prompt for the "Ideological (IL)" Category

English translation of the classification prompt (External / EXT):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "External (EXT)"?

Definition of External (EXT):
A passage is classified as External if the measure is motivated by external economic pressures, trade obligations, or international agreements.

This includes:

- Measures adopted in response to balance-of-payments problems, exchange rate pressures, or the need to protect reserves.
- Tax modifications explicitly linked to supporting exports, foreign trade operations, or competitiveness in international markets.
- Adjustments mandated by commitments to international organizations or treaties (e.g., WTO, EU, EFTA).
- Policy changes justified by concerns about investor confidence or market reactions to the government's financing needs.

*Do **not** classify as External if:*

- The motivation is purely domestic (e.g., deficit reduction, consolidation, demand management, social equity, environment).
- The change is a routine adjustment unrelated to trade, finance, or international agreements.
- The measure concerns only domestic consumption, revenue, or distribution without an external framing.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 9: Classification Prompt for the "External (EXT)" Category

English translation of the classification prompt (Deficit Consolidation / DC):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label "Deficit Consolidation (DC)"?

Definition of Deficit Consolidation (DC):
A passage is classified as Deficit Consolidation if the measure is motivated by a structural or medium- to long-term strategy aimed at strengthening public finances or reducing deficits, borrowing, or public debt.

This includes:

- Measures presented as part of a sustained fiscal plan to ensure the soundness of public finances.
- Structural revenue generators (e.g., base broadening, duty escalators, restrictions on reliefs) intended to improve the budget balance over several years.
- Policies explicitly linked to fiscal consolidation, fiscal sustainability, or medium-term deficit reduction, even when other justifications are also mentioned.
- Measures whose initial intent was deficit consolidation, even if the rhetoric later evolved.

*Do **not** classify as Deficit Consolidation if:*

- The motivation concerns short-term revenue needs ("to raise additional revenue this year," "to finance support for the economy")—this falls under Deficit Reduction (DR), not Consolidation.
- The change is justified mainly by environmental, health, or social equity objectives without a fiscal framing.
- The measure reflects routine indexation or neutral adjustments without a consolidation goal.
- The policy reduces taxes or increases spending without compensating measures.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
  - Yes — [short rationale]
  - No — [short rationale]

Prompt 10: Classification Prompt for the "Deficit Consolidation (DC)" Category

E Replication of Romer and Romer

E.1 Probability Benchmark Settings

E.1.1 Narrative extraction

This appendix documents the benchmark implementation behind Section 5.1. The edited RR narratives are used to validate the probability pipeline rather than a direct binary relabeling exercise. Each example below shows the portion removed before the narrative is passed to the summarization stage. As in the main text, the deleted block includes the explicit RR classification sentence and the two immediately preceding sentences. Removing these passages prevents the model from seeing the authors' bottom-line judgment while leaving enough narrative detail for the summarizer to recover multiple stated motivations from the remaining text.

Example 1: an endogenous measure

The motivation for the Revenue Act of 1951 was again the increase in spending related to the Korean War. The *1951 Economic Report* stated: "These new taxes are required to finance the defense effort; and to help keep total spending within the capacity of current production, so that inflation does not reduce the purchasing power" (p. 17). This same sentiment was echoed in a number of presidential speeches. On February 2, 1951, Truman said, "If we do not tax ourselves enough to pay for defense expenditures, the Government will . . . add to total purchasing power and inflationary pressures" (*Special Message to the Congress Recommending a "Pay as We Go" Tax Program*, p. 2). These quotations make it clear that the tax increase was designed to keep growth normal.

It stated: "The military action in Korea, coupled with the general threat to world peace, has made it necessary to provide extraordinary increases in revenues to meet essential national defense expenditures" (82d Congress, 1st Session, House of Representatives Report No. 586, 6/18/51, p. 1). It also stressed the temporary nature of the tax, saying: "The $12\frac{1}{2}$-percent flat across-the-board increase [in individual income tax rates] was selected as the form of increase, since it is not intended that this increase will be permanent, and therefore, it was desired to provide an increase which would not become an integral part of the rate structure" (p. 8). Because the tax increase was designed to counteract the effects of higher military spending and thereby keep growth normal, I classify it as an endogenous, spending-driven action.

Truman's speech upon signing the bill stated that it would raise about $5.5 billion of additional revenues at an annual rate (*Statement by the President Upon Signing the Revenue Act of 1951*, 10/20/51, p. 1). Both the *1952 Economic Report* (p. 135) and the *1951 Treasury Annual Report* (p. 44) gave the figure as $5.4 billion. I follow my usual practice and use the number from the *Economic Report*.

The increase in corporate taxes was retroactive to April 1, 1951, while the remainder took effect on November 1, 1951 (*1951 Treasury Annual Report*, p. 51). Because the action took place before the middle of the quarter, I date the tax increase in 1951Q4. Approximately $2.3 billion of the tax increase was from corporate taxes (p. 501). Therefore, the retroactive portion of the bill increased taxes at an annual rate of $2.3 billion · 2 extra quarters, or $4.6 billion. Adding this to the steady-state effect of $5.4 billion yields a tax increase of $10 billion in 1951Q4. The return to the steady-state level of tax increase implies a tax cut of $4.6 billion in 1952Q1. If one ignored the retroactive part of the legislation, there would be just a tax increase of $5.4 billion in 1951Q4.

The tax increase largely took the form of an increase in marginal rates. The act also raised the capital gains tax, the tax on corporate profits, and some excise taxes (*1951 Treasury Annual Report*, pp. 50–52).

Most of the changes were explicitly temporary. The individual income tax increases were legislated to expire on January 1, 1954; the corporate and excise tax increases were to continue until March 31, 1954 (*1951 Treasury Annual Report*, pp. 51–52).

Example 2: an exogenous measure

This bill was a significant change of the income tax system. It was the first comprehensive revision of the Internal Revenue Code in 75 years (*Statement by the President Upon Signing Bill Revising the Internal Revenue Code*, 8/16/54, p. 1).

This bill was passed primarily to increase the fairness and efficiency of the tax system. It had been under discussion since 1952 (Holmans, 1961, p. 219), well before the 1953–54 recession. At the signing ceremony, Eisenhower stressed: "It is a law which will help millions of Americans by giving them fairer tax treatment than they now receive." He also said, "economic growth will be fostered by such provisions as more flexible depreciation and better tax treatment of research and development costs, thus encouraging all business—large and small—to modernize and expand" (*Statement by the President Upon Signing Bill Revising the Internal Revenue Code*, 8/16/54, p. 1). The Report of the Committee on Ways and Means to Accompany H.R. 8300 (Internal Revenue Code of 1954) repeated this motivation. It stated: "In general, the purpose of these changes has been to remove inequities, to end harassment of the taxpayer and to reduce tax barriers to future expansion of production and employment" (83d Congress, 2d Session, House of Representatives Report No. 1337, 3/9/54, p. 1).

From this perspective, the reduction in revenues at a given level of income was the result of a belief that the bill would stimulate growth and of a desire to make the reform roughly revenue neutral after accounting for the growth effects. The *1954 Economic Report* stated, "A number of additional tax measures should now be taken in order to strengthen the forces of growth in employment and production. These measures, which involve some immediate sacrifices of revenue, contain the seeds of important future revenue gains to be reaped from the economic growth they will stimulate" (p. 79). The *1955 Economic Report* described the motivations for the bill as "to stimulate competitive enterprise, to strengthen the floor of security for the individual, and to curb tendencies toward either depression or inflation," and referred to "the loss of 1.4 billion dollars as a result of enacting structural tax changes" as almost a side effect of the reform (pp. 19–20).

The possible countercyclical benefits of the bill were certainly mentioned frequently. As discussed above, the administration favored some additional tax relief in mid-1954 because of further declines in defense spending. And, much of the Congressional debate focused on the need for additional stimulus. Indeed, several amendments were offered that involved increasing the personal exemption and cutting tax rates (Holmans, 1961, pp. 229–233). However, once the Excise Tax Reduction Act of 1954 was passed, the Administration argued strenuously against further tax cuts, especially the increase in the personal exemption.

[1] On March 15, 1954, Eisenhower said: "at this time economic conditions do not call for an emergency program that would justify larger Federal deficits and further inflation through large additional tax reductions" (*Radio and Television Address to the American People on the Tax Program*, 3/15/54, p. 4; see also *Annual Budget Message to the Congress: Fiscal Year 1955*, 1/21/54, p. 8). Also, the House report on the bill explicitly downplayed the countercyclical aspects, saying: "This bill is a long overdue reform measure which is vitally necessary regardless of the momentary economic conditions and should not be confused with other measures which may be, or might become, appropriate in the light of a particular

short-run situation" (House Report No. 1337, pp. 1–2). Thus, it seems clear that at least some of the tax cuts contained in the Excise Tax Reduction Act of 1954 and the Internal Revenue Code of 1954 should be classified as exogenous.

[2] Thus, it seems clear that at least some of the tax cuts contained in the Excise Tax Reduction Act of 1954 and the Internal Revenue Code of 1954 should be classified as exogenous. There was a cut beyond what the administration thought was justified by the decline in spending and the state of the economy. Because the excise tax reduction came first and had no obvious other motivation, I choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action.

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[4] Because the excise tax reduction came first and had no obvious other motivation, we choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action. We feel this makes sense given that the Internal Revenue Code of 1954 clearly had a substantial exogenous component and indeed was passed after it was widely understood that the trough of the recession (May 30, 1954) had been turned (Holmans, 1961, p. 233). But, at some level, which of the two tax cuts is classified as endogenous and which as exogenous is somewhat arbitrary.

The act was signed in mid-August 1954, so I date the initial tax change in 1954Q3. My sources give a figure of –$1.4 billion for the revenue effects at an annual rate (for example, *1955 Economic Report*, p. 20; *1954 Treasury Annual Report*, p. 44). The bill was retroactive to January 1, 1954 (*1954 Treasury Annual Report*, p. 286). It thus reduced taxes by $3 \cdot \$1.4$ billion, or $4.2 billion at an annual rate, in 1954Q3. The return to the steady-state reduction of $1.4 billion in 1954Q4 implies a tax increase of $2.8 billion in 1954Q4. If one chose to neglect the retroactive element, there would be just a tax cut of $1.4 billion in 1954Q3.

The changes consisted largely of the removal of obvious inequities, complications, ambiguities, and opportunities for tax avoidance, and of provisions designed to stimulate investment. All the changes were legislated to be permanent.

E.1.2 Probability-benchmark prompts

The main RR benchmark mirrors the workflow described in Section 2. First, the edited narrative is summarized into distinct motivations. Second, each extracted motivation is scored against the narrative taxonomy, and those motivation-level outputs are fused and aggregated into the policy-level structural probability `p_structural`.

The resulting implementation-specific category probabilities are fused and then aggregated into the policy-level `p_structural` exactly as in Section 2; see equations (**??**)–(**??**). The RR benchmark therefore reuses the same probability object as the main application rather than introducing a separate RR-specific aggregation rule.

E.1.3 Legacy binary benchmark

For comparison with earlier drafts, I also retain a legacy benchmark in which each edited RR narrative is mapped directly into a binary exogenous/endogenous label. This direct classification exercise is no longer the main benchmark in the paper, but it remains useful for robustness and for comparing the

You are given an edited Romer–Romer narrative for one tax episode. Produce a structured summary that preserves each distinct stated motivation for the tax change.

- Keep separate motivations separate.
- Do not collapse conflicting motives into a single dominant label.
- Use only motivations supported by the text that remains after editing.

Return a JSON object with:

- `policy_context`: a short description of the policy episode
- `motivations`: a list of short motivation statements

Prompt 11: Structured Summarization Prompt for Edited RR Narratives

You are given one extracted motivation from an edited Romer–Romer narrative. Assess how strongly this motivation supports each narrative category used in the SANI taxonomy.

- Evaluate the motivation category by category.
- Do not force a single dominant label when the evidence is mixed.
- Briefly justify the categories that receive the strongest support.

Return a JSON object with:

- `category_scores`: category-level support scores
- `justification`: a short textual rationale

Prompt 12: Motivation-Level Scoring Prompt for Edited RR Narratives

probability pipeline to the earlier binary design.

Table 6: Legacy Binary RR Classification Performance Summary

LLM Model	False Endog. (%)	False Exog. (%)	Overall Accuracy (%)
GPT-4o-mini (Closed)	31%	4%	82%
Gemini-2.0-flash (Closed)	35%	0%	82%
Llama3.2:4B (Open)	54%	4%	70%

Note: This table reports the earlier direct binary benchmark in which an edited RR narrative was mapped straight into an exogenous/endogenous label without the intermediate motivation-level probability pipeline.

Legacy binary prompt structure. The legacy classification task is implemented using a structured prompt composed of three components: an explicit persona instruction (Prompt 13), a set of base classification instructions mirroring the criteria developed in Romer and Romer (2010) and Cloyne (2013) (Prompt 14), and a small number of illustrative examples implementing a few-shot learning approach (Prompt 15). Drawing on recent work in computer science on prompt engineering to improve performance (White et al., 2023; Chen et al., 2024), this structure is designed to elicit consistent and transparent binary classifications while remaining fully replicable.

> You are a professional economist specializing in Fiscal Policy and Causal Inference. You are highly skilled at identifying whether tax policy changes are endogenous or exogenous, based on economic intent and context.

Prompt 13: Legacy Binary Persona Prompt

> You will be provided descriptions of legislated tax changes in the U.S. from 1945 to 2007. Please classify each tax change as either endogenous (N) or exogenous (X) based on the following definitions:
>
> - **Endogenous (N)**: If the primary motivation for the tax change was related to addressing current or expected economic conditions — e.g., reacting to a recession, inflation, unemployment, or funding new programs.
> - **Exogenous (X)**: If the tax change was mainly introduced to deal with structural or long-term issues — e.g., reducing an existing deficit, encouraging growth, improving fairness or efficiency, or philosophical shifts in policy. These actions should not be tied to short-term economic fluctuations.
>
> **Guidance**: If there are multiple motivations, use the predominant one cited at the time of passage.
>
> Your response must ALWAYS be a JSON object containing two fields:
>
> - `classification` (string, either N or X)
> - `justification` (a brief explanation)

Prompt 14: Legacy Binary Base Instruction Prompt

Endogenous

- Responds to contemporaneous or forecasted macro conditions (recession, inflation, unemployment)? Yes→N
- Primarily to finance newly enacted or announced spending? Yes→N
- Focus on short-term stabilization rather than long-term reform? Yes→N

Exogenous

- Primarily to reduce a non-program deficit, not used as a stabilization tool? Yes→X
- Chiefly to promote long-run growth, equity, efficiency, or philosophical changes? Yes→X
- Intended to raise long-run potential output or reduce structural unemployment without reacting to a current shock? Yes→X

Prompt 15: Legacy Binary Few-Shot Prompt

E.2 Legacy Binary Benchmark Robustness checks

E.2.1 Training-data leakage robustness check

To assess whether the language model has memorized specific source material, I implement a sentence-reconstruction test taken from Golchin and Surdeanu's (2024) contamination detection framework. The intuition is the following: if a model has encountered a document during training, it may be able to reproduce missing content from that document with unusual precision—especially when provided with cues pointing to the original source. To implement this test, I systematically remove sentences from the Romer and Romer narratives and asks the model to reconstruct them under two conditions: one relying solely on the surrounding context, and the other explicitly guiding the model by referencing the source paper and section. If the model performs significantly better under guided prompting, this may suggest prior exposure to the original text. To quantify and evaluate the reconstruction quality, I use both semantic (embedding cosine similarity) and lexical (ROUGE-L) metrics, followed by a bootstrap-based significance test to flag cases of potential memorization.

Using both metrics, and evaluating with `gpt-4o-mini` and `gemini-2.0-flash`, no narrative is flagged as a potential case of memorization. This result strengthens confidence that the model was not trained on Romer and Romer (2010), and supports the validity of the replication exercise by reducing concerns about training-data leakage as a source of bias in the model's classifications.

Procedure.

1. **Multiple draws & prompts.** For each narrative d I perform $T = 10$ draws. In draw t, one sentence $s_{d,t}$ is chosen uniformly at random, removed, and replaced by the token `[MISSING SENTENCE]`. I then issue two prompts a general prompt, and a guided prompt (Prompts 16).

   The model returns two reconstructions $\hat{s}^{\text{gen}}_{d,t}$ and $\hat{s}^{\text{guide}}_{d,t}$.

2. **Similarity metrics.** For each original $s_{d,t}$ and each reconstruction, compute:

   - *Embedding cosine similarity* $\rho^{\text{gen}}_{d,t} = \cos\big(s_{d,t}, \hat{s}^{\text{gen}}_{d,t}\big)$, $\rho^{\text{guide}}_{d,t} = \cos\big(s_{d,t}, \hat{s}^{\text{guide}}_{d,t}\big)$, using `text-embedding-3-large`.
   - *ROUGE-L F1 score* $r^{\text{gen}}_{d,t}$ and $r^{\text{guide}}_{d,t}$ between the original and each reconstruction.

General prompt:
You are provided with a passage in which one sentence has been removed and replaced with `[MISSING SENTENCE]`. Based ONLY on the context provided, generate a plausible sentence that could fill the gap. You do not need to match any specific source text — just ensure the result is coherent and contextually appropriate.

{`narrative text from Romer and Romer`}

Reconstructed sentence:

Guided prompt:
You are provided with a passage from the scientific paper *A Narrative Analysis of Postwar Tax Changes* by Christina D. Romer and David H. Romer, published in 2010. The passage comes from the section titled '{`title`}" in the main body of the paper.
One sentence has been removed and replaced with `[MISSING SENTENCE]`. Based **only** on the context provided and your knowledge of this specific paper, reconstruct the missing sentence exactly as it appeared.

{`narrative text from Romer and Romer`}

Reconstructed sentence:

Prompt 16: General and Guided Prompt

3. **Paired bootstrap test.** For each narrative d, define the mean differences

$$\Delta\rho_d \;=\; \frac{1}{T}\sum_{t=1}^{T}\big(\rho_{d,t}^{\text{guide}} - \rho_{d,t}^{\text{gen}}\big), \quad \Delta r_d \;=\; \frac{1}{T}\sum_{t=1}^{T}\big(r_{d,t}^{\text{guide}} - r_{d,t}^{\text{gen}}\big).$$

We perform a nonparametric paired bootstrap with $B = 1{,}000$ resamples of the T pairs $\{(\rho_{d,t}^{\text{guide}}, \rho_{d,t}^{\text{gen}})\}$ (and similarly for $\{(r_{d,t}^{\text{guide}}, r_{d,t}^{\text{gen}})\}$), yielding bootstrap distributions of $\Delta\rho_d$ and Δr_d. Each indicator $\mathbf{1}\{\Delta\rho_d^{*(b)} \le 0\}$ (or $\mathbf{1}\{\Delta r_d^{*(b)} \le 0\}$) counts a bootstrap sample in which the similarity under the guided prompt is no better than—or worse than—that under the general prompt; that is, the model failed to improve reconstruction quality despite being explicitly cued about the source. The empirical p-values are

$$p_d^{\rho} \;=\; \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\{\Delta\rho_d^{*(b)} \le 0\}, \quad p_d^{r} \;=\; \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\{\Delta r_d^{*(b)} \le 0\}.$$

Decision rule. A narrative d is flagged as potentially memorized if its bootstrap p-value for either similarity metric falls below 0.05, i.e. if guided prompting produces significantly closer reconstructions than general prompting.

E.2.2 Manually Investigating Legacy Binary Misclassifications

The misclassification pattern, mostly exogenous policies misclassified as endogenous, can possibly be attributed to an asymmetry inherent in macroeconomic narrative analysis: while it is relatively straightforward to construct narratives supporting the endogeneity of a policy intervention, establishing

its exogeneity typically requires more stringent evidence. In the absence of sufficiently strong signals of such independence LLMs may default to a conservative interpretation, erring on the side of endogeneity.

Steering growth above normal levels. A recurring source of systematic misclassification involves tax cuts enacted with the aim of boosting growth above normal levels. In at least three instances, the LLM interprets these as countercyclical, and thus endogenous, due to the presence of growth-oriented motivation. However, according to Romer and Romer's decision rule, such motivations qualify as exogenous, as they are not designed to return output to trend but to exceed it. This kind of mislabeling can be mitigated by explicitly including this rule in the prompt design or few-shot examples, highlighting that aspirational growth policies, unless tied to negative output gaps, are to be treated as exogenous.

Conflicting motivations. Another set of misclassifications arises from cases with mixed or conflicting motivations. For example, one narrative discusses a tax increase intended primarily for deficit reduction and solvency of the Highway Trust Fund, both exogenous motives, but also references inflation concerns and contemporaneous spending increases. While the latter suggest endogenous intent, the original classification deems the measure exogenous with noted ambiguity. Such nuanced judgment is difficult for LLMs to replicate without additional guidance on prioritizing primary over secondary motivations.

"Package" related measure. Finally, there are highly specific edge cases that hinge on bespoke decision rules. A salient example is the Tax Adjustment Act of 1966. Despite being implemented to counteract high growth following an exogenous tax cut, Romer and Romer classify it as exogenous, arguing that treating it as endogenous would yield nonsensical results. This type of reasoning, context-specific and precedent-based, poses a significant challenge for LLMs, particularly in the absence of similar examples in the prompt.

E.3 Legacy Binary Benchmark: Confidence Diagnostics

This subsection reports confidence diagnostics for the legacy binary benchmark. A model that consistently produces accurate direct classifications but is highly uncertain about its judgments might be less reliable in practice, especially in ambiguous cases. To gauge this, I examine two key metrics: the log probabilities of generated tokens and the perplexity of the overall output.

Log probabilities / ambiguity. The log probabilities of generated tokens provide a direct measure of the model's certainty at each step of its output. When the LLM outputs a classification (e.g., "X" for exogenous or "N" for endogenous), each token in that word has an associated log probability. A higher (less negative) log probability indicates greater confidence in that specific token's prediction. By aggregating these values, I can assess the model's conviction in its final classification. For instance, a significantly lower log probability for the chosen classification in a misclassified case, compared to a correctly classified one, would suggest that the model was less certain in its incorrect judgment.

To quantify this uncertainty more systematically, I define an *ambiguity* score based on the model-assigned probability for the predicted class label. Specifically, for a classification such as "X", I extract the associated probability p from the model output and compute ambiguity as $\min(p, 1-p)$. This

transformation maps probabilities near 0 or 1 (high confidence) to values near 0 (low ambiguity), and probabilities near 0.5 (low confidence) to values near 0.5 (high ambiguity). This score allows for a consistent interpretation of how uncertain the model is about its decision, regardless of whether the predicted label is "X" or "N".

Perplexity. In addition to token-level confidence, I evaluate the perplexity of the overall output, including any explanatory text or justification provided by the LLM alongside its classification. Perplexity is a measure of how well a probability model predicts a sample. A lower perplexity score indicates that the model is "less surprised" by its own output, suggesting a more coherent and confident generation. Conversely, a high perplexity implies greater uncertainty or hesitation, potentially indicating that the model struggled to formulate a clear rationale or was less sure about its classification.

Relation between model confidence and misclassification. To empirically assess whether these uncertainty measures predict classification errors, I fit three logistic regression models where the binary outcome variable is whether the classification was incorrect. The first model includes ambiguity (based on the model's token-level output probabilities), the second includes perplexity, and the third includes both predictors. The results, shown in Table 7, indicate that ambiguity is a strong and statistically significant predictor of error: higher ambiguity is associated with a significantly increased likelihood of misclassification. Specifically, in the full model, the coefficient on ambiguity is 4.26, which implies that a one-unit increase in ambiguity multiplies the odds of misclassification by more than 70 times (since $e^{4.26} \approx 70.7$). This is a large effect, underscoring the critical role of ambiguity in driving model errors.

In contrast, the effect of perplexity—while statistically significant when ambiguity is excluded—disappears once ambiguity is included. In the joint model, the coefficient on perplexity becomes small and non-significant, suggesting that ambiguity subsumes most of the predictive value of perplexity. In other words, while perplexity correlates with misclassification on its own, it does not offer additional explanatory power beyond what is already captured by token-level ambiguity.

Table 7: Logistic Regression of Error on Ambiguity and Perplexity

	Ambiguity	**Perplexity**	**Ambig+Perp**
const	-1.358***	-3.464***	-1.068
	(0.085)	(0.454)	(0.788)
ambig	3.893***	—	4.258***
	(0.606)		(1.159)
perp.	—	2.187***	-0.292
		(0.417)	(0.787)
N	1000	1000	1000
Pseudo R^2	0.037	0.025	0.037

Note: Coefficients are log-odds estimates; standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A dash (—) indicates the variable was omitted from that model.

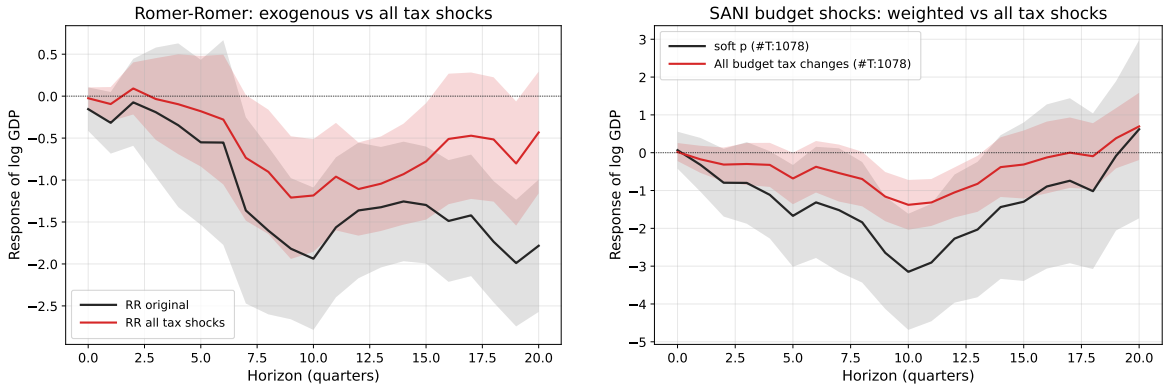
E.4 The Role of Narrative Exogeneity in Tax Shock Identification

This subsection jointly examines the role of narrative exogeneity in the identification of tax shocks under the Romer–Romer (2010) framework and the SANI procedure. Figure 11 reports impulse responses obtained using, for each approach, (i) all legislated tax changes without conditioning on policy motivations, and (ii) tax shocks constructed using narrative-based exogeneity restrictions.

Across both frameworks, the results point to a common pattern. Including all tax changes without narrative filtering leads to a substantial attenuation of the estimated output response to a tax increase. In the Romer–Romer case, peak effects are reduced by roughly one third relative to the baseline specification that isolates exogenous shocks, while preserving a broadly similar hump-shaped dynamic profile. In the SANI case, the attenuation is even more pronounced: treating all routine budgetary tax measures as exogenous yields responses that are markedly smaller in magnitude than those implied by narrative-weighted shocks.

In both panels, confidence bands are mechanically tighter when all tax changes are included. This increase in precision reflects the inclusion of endogenous policy variation rather than improved identification. Conditioning on stated policy motivations, whether through discrete narrative classification or continuous exogeneity weighting, is therefore essential to recover economically meaningful tax shocks.

Taken together, the side-by-side comparison highlights the central role of narrative information in fiscal shock identification. While the institutional settings and policy universes differ across the two approaches, the qualitative implications are strikingly similar: absent narrative exogeneity restrictions, estimated tax multipliers are severely attenuated, despite appearing more precisely estimated. This pattern underscores the importance of narrative methods for isolating exogenous fiscal variation in the presence of policy endogeneity.



(a) Romer–Romer: Exogenous vs. All Tax Shocks

(b) SANI: Narrative-Weighted vs. All Tax Shocks

Figure 11: The Role of Narrative Exogeneity in Tax Shock Identification

Note: Each panel reports impulse responses of real GDP to a one-percent tax increase estimated using the same empirical specification. The black curve corresponds to shocks constructed using narrative exogeneity restrictions (Romer–Romer exogenous shocks in the left panel and SANI narrative-weighted shocks in the right panel). The red curve reports responses obtained when all legislated tax changes (endogenous included), without narrative filtering. Shaded areas denote 68% confidence intervals.

Augmenting the RR benchmark with budget-based SANI shocks. As an additional robustness check, I augment the original Romer–Romer shock series with budget-based SANI shocks in quarters not covered by the original RR narrative instrument. The goal is not to redefine the benchmark, but to test whether the additional documentary variation mainly adds support or materially shifts the estimated response.

Figure 12 shows that the augmented series tracks the original RR response closely. Point estimates remain similar across horizons, while confidence bands narrow modestly at short and medium horizons. In that sense, the additional SANI-identified budget measures operate primarily as a support-expansion device rather than as a qualitatively different source of identification.

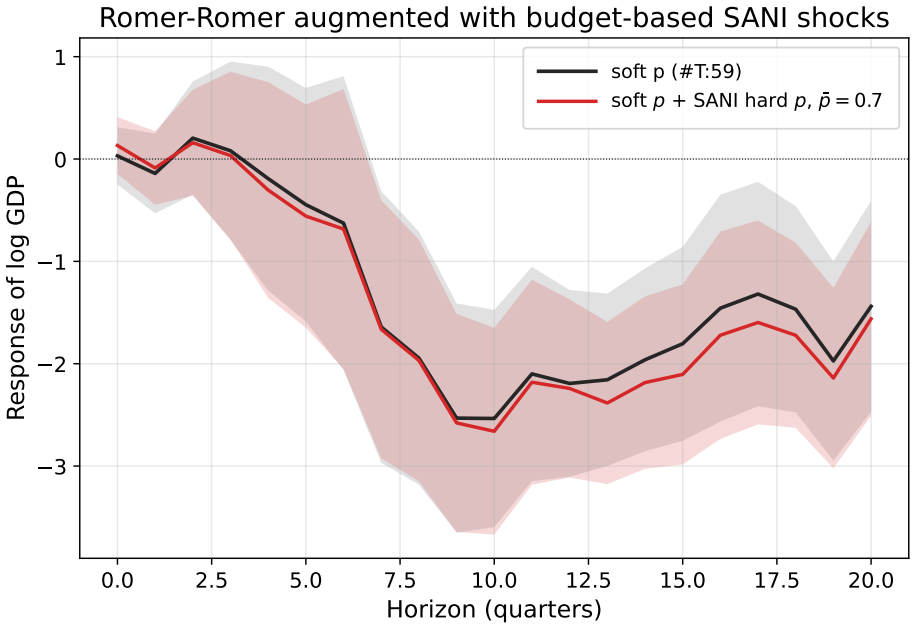


Figure 12: Impulse Responses to Tax Shocks: Romer–Romer vs. Romer–Romer Augmented with SANI

Note: The black line reports impulse responses obtained using the original Romer–Romer exogenous tax shock series. The red line corresponds to a combined series that augments the Romer–Romer shocks with exogenous tax measures identified by SANI outside the original narrative sample. Shaded areas denote 68% confidence intervals.

F Other

F.1 VAT table adoption

Table 8: VAT (or GST-equivalent) Adoption by Year in Developed Economies

Year	**Countries**
1954	France
1967	Denmark
1968	Germany
1969	Netherlands; Sweden
1970	Luxembourg; Norway
1971	Belgium
1972	Ireland
1973	Austria; Italy; United Kingdom
1977	South Korea
1980	Mexico
1985	Turkey
1986	New Zealand
1989	Japan
1990	Iceland
1991	Canada
1994	Finland
1995	Switzerland
2000	Australia